Draft version: May 3, 2023. Supplemental materials are included at the end of the document

This is a preprint of an Open Access article published in *Collabra: Psychology*

Coll-Martín, T., Carretero-Dios, H., & Lupiáñez, J. (2024). Attention-deficit/hyperactivity disorder symptoms as a function of arousal and executive vigilance: Testing Halperin and Schulz's neurodevelopmental model in a sample of community adults. *Collabra: Psychology*, *10*(1), Article 121392. https://doi.org/10.1525/collabra.121392

Attention-Deficit/Hyperactivity Disorder Symptoms as Function of Arousal and Executive Vigilance: Testing the Halperin and Schulz's Neurodevelopmental Model in an Adult Community Sample

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The design and analysis plan of this study were preregistered in the Open Science Framework on July 10, 2022 using the Preregistration for Quantitative Research in Psychology Template (PRP-QUANT; Bosnjak et al., 2022): https://osf.io/tkdq7. Data and material, including R scripts, are publicly available at https://osf.io/vqgms/. The online cognitive task employed in this study (ANTI-Vea) is freely available at https://anti-vea.ugr.es/. We have no potential conflict of interests to disclose. Our work was supported by a predoctoral fellowship (FPU17/06169) awarded to Tao Coll-Martín from the Spanish Ministry of Education, Culture, and Sport; a research project grant (PID2019-104239GB-I00) from the Spanish Ministry of Science, Innovation and University (State Research Agency/10.13039/501100011033) awarded to Hugo Carretero-Dios; and research project grants by Spanish Ministry of Economy, Industry and Competitiveness, MCIN/ AEI /10.13039/501100011033 (PID2020.114790GB.I00), and PY20_00693, funded by the Consejería de Universidad, Investigación e Innovación, Junta de Andalucía, and by FEDER A way of doing Europe, awarded to Juan Lupiáñez. The funders had no role in any stage of the development and publication of this work. This paper is part of the doctoral dissertation of the first author under the supervision of the last two authors. We thank Rocío Martínez Caballero for her generous availability and invaluable support with the development of the online version of the task.

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Abstract

Halperin and Schulz's neurodevelopmental model postulates that the onset of attention-deficit/hyperactivity disorder (ADHD) in childhood is due to subcortical alterations, whereas the disorder trajectory into adulthood depends on the development of executive functions. Based on a dimensional framework of ADHD, Coll-Martín et al. (2021) found support for the model in an adult community sample assessed in arousal and executive vigilance. The present study is a preregistered (https://osf.io/tkdq7) close replication of Coll-Martín et al. with stricter control of statistical error rates to test the two hypotheses of the model. A sample of university students (*N* = 292 valid; 49% women) completed self-reports of ADHD symptoms in childhood (retrospectively) and adulthood and performed the online version of an attentional task (the ANTI-Vea). Our preregistered hypotheses achieved an acceptable statistical power for the effects of interest, even after accounting for measurement error. Despite this, only the unexpected negative correlation between executive vigilance and symptoms in childhood was significant, therefore refuting the theoretical predictions. Similarly, neither multiverse nor exploratory analyses supported the dissociation pattern proposed by the neurodevelopmental model. ADHD symptoms across the lifespan may be pathophysiologically identical, at least in terms of vigilance. Future studies could include complementary assessment methods and clinical groups.

Keywords: attention-deficit/hyperactivity disorder, arousal vigilance, executive vigilance, neurodevelopment, community adults

Attention-Deficit/Hyperactivity Disorder Symptoms as Function of Arousal and Executive Vigilance: Testing the Halperin and Schulz's Neurodevelopmental Model in an Adult Community Sample

Attention-deficit/hyperactivity disorder (ADHD) is a neurodevelopmental condition characterized by age-inappropriate, persistent, and impairing levels of inattention and/or hyperactivity–impulsivity (American Psychiatric Association [APA], 2013). The disorder is present in about 5% of children and 2.5% of adults (Polanczyk et al., 2007; Simon et al., 2009). Throughout the lifespan, ADHD is a risk factor for several negative outcomes, including educational underachievement, difficulties with employment, and criminality (Faraone et al., 2015; Fletcher, 2014; Loe & Feldman, 2007). Against this backdrop, identifying the neurocognitive mechanisms underlying ADHD symptoms across development is crucial to enhance the approach to the disorder (Castellanos & Tannock, 2002; Luo et al., 2019; Sonuga-Barke et al., 2023).

Neurodevelopmental Model of ADHD

Numerous theoretical models have formulated different explanations of ADHD based on underlying impairments in single (Barkley, 1997; Sergeant, 2005; Sonuga-Barke et al., 1992) or multiple (Durston et al., 2011; Sonuga-Barke, 2003) neurocognitive pathways. In an attempt to account for the developmental trajectory of the disorder, Halperin and Schulz (2006) elaborated a *neurodevelopmental model of ADHD* (also referred here as "neurodevelopmental model"). The authors postulate a double dissociation in which the onset of ADHD in childhood is due to subcortical brain dysfunctions that remain stable over the lifespan, while the reduction of symptom severity with age is dependent on the development of the prefrontal cortex. In this

sense, executive functions, mediated by the prefrontal cortex, are not central to the early emergency of ADHD, but influence on its developmental course into adulthood.

The neurodevelopmental model generates testable predictions in measures of cognitive tasks. First, the subcortical brain dysfunctions should be observable in measures with minimal or no executive load, such as basic reaction time (RT) or RT variability in simple RT task. These measures should distinguish ADHD children from controls across the lifespan, regardless of adolescent or adult status. Second, measures of executive function, such as target identification or inhibition tasks, should be dimensionally related to the severity of ADHD symptomatology, particularly in adulthood. Importantly, the model considers the phenomenon of late-onset ADHD (Asherson & Agnew-Blais, 2019), which would be product of either early or late lesions in the prefrontal cortex subserving executive processes.

Initial support for the neurodevelopmental model came from a prospective longitudinal study of 98 children with ADHD who were reassessed after 10 years (Halperin et al., 2008). Compared to matched controls, the adolescents diagnosed with ADHD in childhood showed deficient response variability, irrespective of their current clinical status. However, only adolescents with persistent—but not remitting—ADHD exhibited poorer inhibition and working memory than controls. Despite the promising findings of this study, subsequent research has yielded mixed results (Coghill et al., 2014; Coll-Martín et al., 2021; Gmehlin et al., 2016; Leenders et al., 2021; McAuley et al., 2014; Rommel et al., 2015). Furthermore, a systematic review found no support for the model, as the patterns of neurocognitive deficits in individuals with ADHD were similar for high- and low-executive-load measures (van Lieshout et al., 2013).

Although the current literature is not very encouraging with the neurodevelopmental model, some gaps and methodological issues are worth considering. First, most of these studies

consisted of case–control comparisons that were underpowered to detect a range of effects that could be considered theoretically relevant (e.g., 0.35 ≥ Cohen's d_s ≥ 0.2). In contrast, population-based or community samples can better fit the dimensional nature of ADHD (Coghill & Sonuga-Barke, 2012; Hilger et al., 2020) and are more efficient in collecting well-powered sample sizes. Second, studies that included low-executive-load outcomes rarely used tasks with no executive component (e.g., Coll-Martín et al., 2021; Gmehlin et al., 2016). In this sense, even RTs or omission errors in tasks such as the continuous performance test (CPT) are influenced by the executive processes and the response criterion involved in the task demands. The next section will address this issue.

Measuring Executive and Arousal Vigilance

Impaired vigilance is central to the phenotypic and neurocognitive characterization of ADHD (Huang-Pollock et al., 2012; Wilding, 2005; Willcutt et al., 2005). The construct is generally defined as the attentional capacity to maintain performance over time. Given the diversity of terms and measures linked to vigilance, some authors deem it as a nonunitary concept (Langner & Eickhoff, 2013; Luna et al., 2018; Sturm et al., 1999). In this vein, Luna et al. distinguish two components of vigilance: *executive vigilance* (EV) and *arousal vigilance* (AV).

EV is the ability to detect infrequent but critical signals among nonsignal stimuli. It is measured with tasks derived from the CPT paradigm such as the AX-CPT (Rosvold et al., 1956) or the Test of Variables of Attention (TOVA; Greenberg & Waldman, 1993). This type of tasks seems to involve executive mechanisms of sustained attention for stimuli discrimination and goal-oriented response selection. Notably, response-inhibition CPTs (e.g., Conners's CPT [Conners, 2000], Sustained Attention to Response Task [SART; Robertson et al., 1997]), the

most common CPT variant in ADHD research, also require motor suppression of the preponderant response in the presence of the target stimulus. Key measures of EV are hits (inverse of omission errors) and false alarms, both tending to decrease with time on task (for a discussion about false alarms, see Thomson et al., 2016).

AV is the capacity to sustain a rapid reactivity to any environmental stimulus—without implementing any control over the selection of the response executed. This form of vigilance is measured in simple RT tasks, such as the Psychomotor Vigilance Test (PVT; Lim & Dinges, 2008) and the WAFA test of the Vienna Test System (Schuhfried, 2013). These tasks seem to record an arousal component of vigilance, a mechanism that could be more related to physiological levels of excitability. The main indices of AV are the mean and variability of the RT and the attentional lapses. Contrary to EV, this vigilance decrement manifests as an increase of the measures during the task.

In order to evaluate both vigilance components simultaneously, Luna et al. (2018) designed the *Attentional Networks Test for Interaction and Vigilance—Executive and Arousal Components* (ANTI-Vea). This task is based on the Attention Networks Test for Interaction (ANTI; Callejas et al., 2004), which combines an Eriksen flanker paradigm with spatial cues and warning signals to assess the three attentional networks (Posner & Petersen, 1990). In ANTI trials, which are the bulk of the ANTI-Vea, participants must respond to the direction pointed by the central arrow. To assess EV, in a small percentage of trials that central arrow appears vertically displaced for participants to detect and respond to it, thereby suppressing their preponderant response to ANTI trials. To measure AV, in another small percentage of trials, a salient stimulus (i.e., a red down counter) is displayed for participants to stop it as fast as

possible. Of note, the length of the task (~33 min) successfully induce a decrement in all vigilance measures (Luna et al., 2018).

Several studies have found that EV and AV are dissociable in the ANTI-Vea, both as a result of experimental manipulations (Feltmate et al., 2020; Hemmerich et al., 2023; Sanchis et al., 2020) and in relation to individual differences (Cásedas et al., 2022; Román-Caballero et al., 2021). In the context of ADHD, Coll-Martín et al. (2021) conducted a study in which we administered the ANTI-Vea[1] to 113 university undergraduates in a laboratory setting. They assessed ADHD symptom severity retrospectively in childhood and concurrently in adulthood through self-reports. In line with the neurodevelopmental model, the authors found that symptoms in childhood correlated with higher lapses (AV), while adult symptoms were associated to a greater decrease in hits during the task (EV). Although promising, these results came from an exploratory study with several outcomes and need to be replicated with a larger sample and a stricter control of false positive rates and some potential confounders.

The Present Study

This study aimed to test the neuropsychological predictions of Halperin and Schulz's (2006) neurodevelopmental model of ADHD from a dimensional framework. For this purpose, a community sample of university students was assessed for childhood and adult symptoms of ADHD and performed the online version of the ANTI-Vea to measure AV and EV. This procedure can be considered a close replication of Coll-Martín et al. (2021), with the main

[1] Technically, both the cited study and the study we are presenting in this manuscript used a version of the ANTI-Vea that incorporates some random ANTI trials with irrelevant distractors. The added trials represented less than 10% of the total, and they did not affect the normal functioning of the rest of the task indices or the reliability scores (Coll-Martín et al., 2021). For the sake of simplicity, in this paper we refer to this version of the task as "ANTI-Vea" and omit details of those additional trials in the description.

difference being the setting where the ANTI-Vea was administered (but see Luna et al., 2021, for the remarkable psychometric similarities between the lab and online versions). Unlike the original study, the larger sample size and the more specific contrasts selected in this study provides a reasonably acceptable informational value to the statistical tests performed.

According to the neurodevelopmental model and the results of the original study (Coll-Martín et al., 2021), we preregistered the following hypotheses (https://osf.io/tkdq7): (H1) childhood ADHD symptoms would predict higher lapses (AV); and (H2) adult ADHD symptoms would predict lower hits (EV), even after accounting for symptoms in childhood (i.e., after controlling for baseline to focus on later development). As primary outcomes, we focused on the measures of vigilance decrement, which has been considered the core feature of the construct (Huang-Pollock et al., 2012; Tucha et al., 2017) and was related to ADHD in both vigilance measures in the original study (Coll-Martín et al., 2021). However, we also calculated the overall vigilance scores as secondary outcomes, since these are the measures typically reported in ADHD research and are more reliable (Cásedas et al., 2022; Coll-Martín et al., 2021; Huang-Pollock et al., 2012).

Method

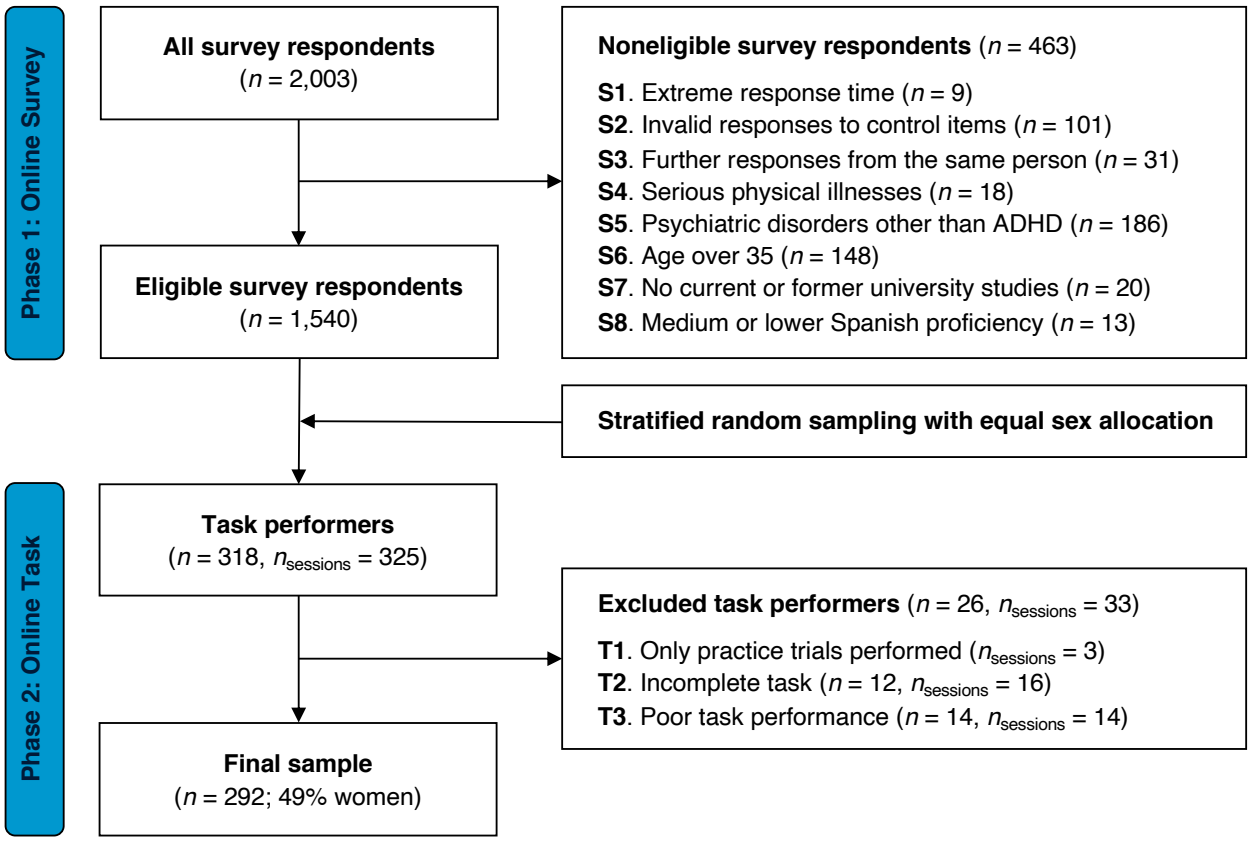
In order to report the severity of the tests transparently, the study design and analysis plan were publicly preregistered with the Preregistration for Quantitative Research in Psychology Template (PRP-QUANT; Bosnjak et al., 2022) at https://osf.io/tkdq7 (for a report of deviations from the preregistration, see Supplemental Table 1). Consequently, the main statistical hypotheses and additional analyses will be referred to with the same label as in the preregistration document.

Sample Selection and Study Procedure

Figure 1 illustrates the sample selection process. In the first phase of the study, we collected a total of 2,003 responses to a 15-min online survey via LimeSurvey (https://www.limesurvey.org). Participants were recruited through advertisements on the virtual distribution list of our university. In exchange for completing the survey, participants had the opportunity to win two prizes of €200 in a raffle. In addition to the self-reports of ADHD symptoms, the survey included sociodemographic questions and other psychological variables (see the preregistration document for further details). The appearance order of the two main self-reports of ADHD symptoms in childhood and adulthood was randomised. After exclusions, there were 1,540 eligible participants (72.7% women, 26.1% men, 1.23% nonbinary; 18–35 years, M = 22.5, SD = 3.7).

Figure 1

Selection Process of Participants Through Each Stage of the Study



Note. S1–S8 are the exclusion criteria of the survey respondents. S1 based on the mean plus/minus three standard deviations cut-off. S2 based on three attention check items distributed throughout the survey. S4–S8 based on the information reported by participants. S4–S5 referred to diagnosed conditions. S4 limited to diseases that may largely impact task performance (e.g., motor paralysis). Note that there were participants qualified for exclusion by more than one survey criteria. T1–T3 are the exclusion criteria of the cognitive task performers. Note that only one task session per participant was included in the final sample.

In the second phase, eligible participants were randomly invited to perform the online version of the cognitive task (i.e., the ANTI-Vea). Invitations were sent via email until

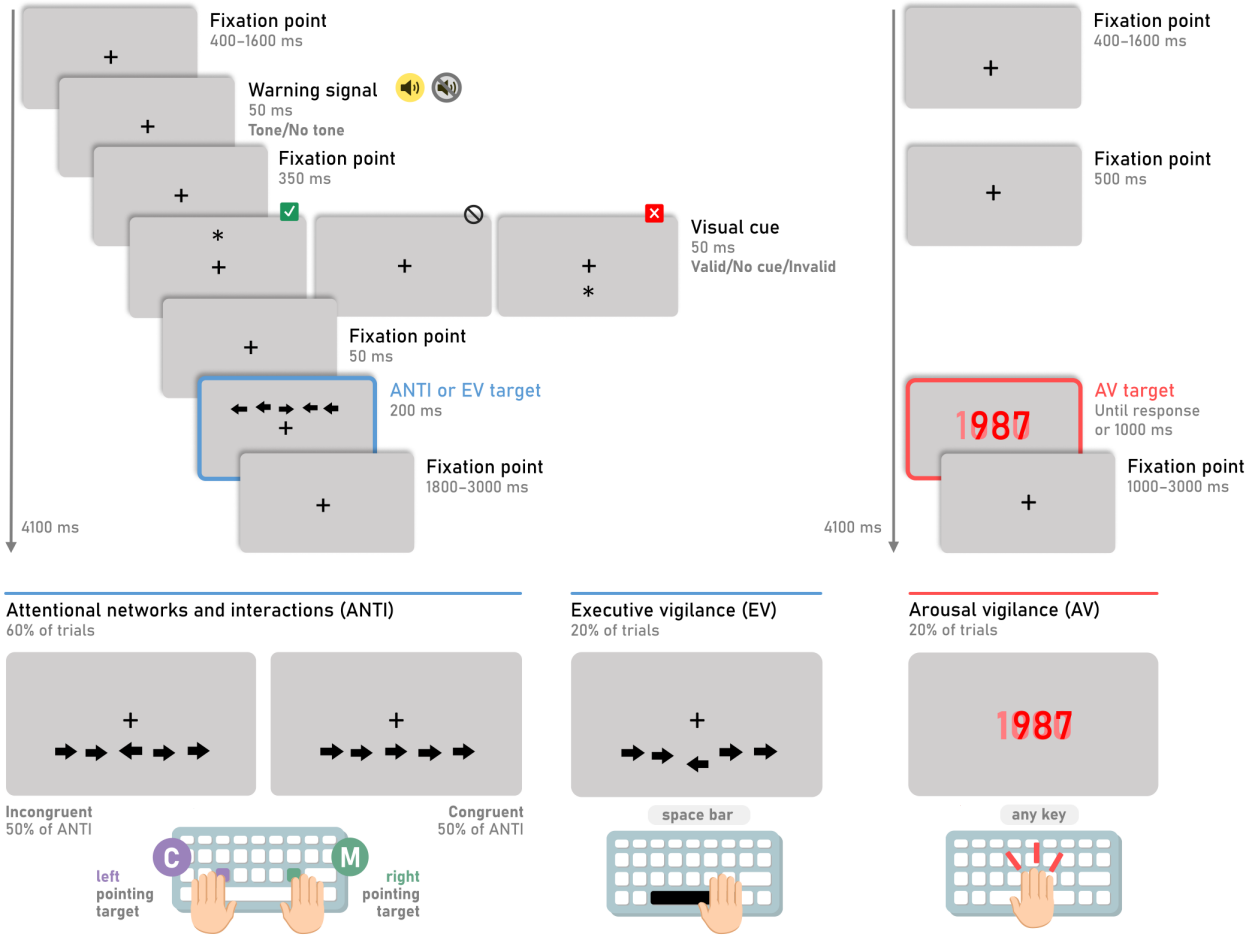
approximately 300 responses were collected. We stratified by sex to ensure a representative distribution of the population in that factor. In addition to the ANTI-Vea, this phase included a brief set of questions about the task (i.e., questions related to the task experience, which are not part of this study, and control items) and a reassessment of ADHD symptoms[2] with the same instruments as in the initial survey. The entire duration of the second phase was about 50 minutes. Participants were compensated €6 for their voluntary participation.

A total of 318 participants (50% women, 50% men) enroled in the second phase of the study. They received the link to the web-based version of the ANTI-Vea with the following instructions before starting the task: (a) sit in a comfortable place without distractions; (b) keep entertainment devices (i.e., television, radio, mobile phone, etc.) out of reach; (c) set the computer's sound level at 75% and do not minimise the automatic full-screen mode of the task; (d) if necessary, wear glasses or contact lenses; and (e) if necessary, solve any particular issue before starting so that the task can be completed without any breaks. The stimuli sequence and correct responses for each type of trial are depicted in Figure 2 (for a more detailed description of the task with audiovisual material, see Method section of the ANTI-Vea website at https://anti-vea.ugr.es/method.html). The ANTI-Vea comprises three types of trials: ANTI (60%), EV (20%), and AV (20%). Participants were encouraged to respond as quickly and accurately as possible while keeping their eyes on the fixation point until the finalization of the task. The ANTI-Vea started with a practice phase, in which instructions and feedback were given so that participants could gradually familiarize themselves with each type of trial.

[2] For 67 participants (17.5% of the final sample), ADHD symptoms were not assessed in the second phase of the study.

Figure 2

Attention Network Test for Interaction and Vigilance—Executive and Arousal Components (ANTI-Vea) Procedure



Note. The top left shows the temporal sequence of the ANTI and EV trials. Target and flankers could appear above or below the fixation cross and point to the left or the right side with equal probability. The warning signal appeared in half of these trials. The visual cue had an equal chance of appearing in the same location as the target, in the opposite location, or not appearing. The bottom left and middle shows the correct response based on whether the target is vertically aligned with the flankers (ANTI trials) or displaced (EV trials). Note that the five arrows are slightly vertically displaced at random by ± 2 px to generate some noise in ANTI and EV trials,

while the target is substantially vertically displaced by ± 8 px in EV trials. The right part shows the temporal sequence and correct response of the AV trials. The duration of each task interval appears next to its corresponding box. Note that, although every trial lasted 4,100 ms, the sequence of events appeared at a variable interval within each trial.

After the practice phase, participants performed the task itself, which consisted of six seamless blocks of 80 pseudorandomised trials each (48 ANTI, 16 EV, and 16 AV). In ANTI trials, a central arrow (i.e., the target) with two flankers on each side appeared pointing to the left or right. Participants had to discriminate the direction of the target (by pressing either "C" key for leftward direction or "M" key for rightward direction) while ignoring the direction of the flanking arrows, which could point to the same or the opposite direction as the target with equal probability. These stimuli could be preceded by an asterisk (i.e., a visual spatial cue) and/or a tone (i.e., a warning signal), thereby allowing the conditions to assess the three attentional networks. EV were similar to ANTI trials, with the exception that here the target appeared vertically displaced from the flankers, so that participants must detect the displacement by pressing the space bar. In contrast, AV trials only displayed a red millisecond down counter (starting at 1,000 ms) for participants to stop it by pressing any key as fast as possible.

All data collection took place from June 2021 to June 2022 entirely online. The research project was approved by our institutional ethics.

Instruments

Barkley Adult ADHD Rating Scale-IV: Childhood and Current Symptoms

The self-reports of the Barkley Adult ADHD Rating Scale-IV (BAARS-IV; Barkley, 2011) include two scales to assess ADHD symptoms: retrospectively in childhood (cBAARS-IV) and concurrently in adulthood (aBAARS-IV). Each scale is composed of 18 items, nine of

inattention (e.g., “Difficulty sustaining my attention in tasks for fun activities”) and nine of hyperactivity–impulsivity (e.g., “Shift around excessively or feel restless or hemmed in”), in a Likert scale ranged from 1 (*never or rarely*) to 4 (*very often*). While the cBAARS-IV refers to behaviours between 5 and 12 years of age, the aBAARS-IV refers to the last 6 months. Since the items are based on the *Diagnostic and Statistical Manual of Mental Disorders* (4th ed.; DSM-IV; APA, 1994), we used the Spanish version of the manual for the translation (APA, 1994/1995), as we did in the previous study (Coll-Martín et al., 2021). In our final sample, Cronbach’s alpha reliability scores were .90 and .86 for cBAARS-IV and aBAARS-IV, respectively, close to the .95 and .92 of the original BAARS-IV (Barkley, 2011). Barkley proposed the 95th percentile of these scales as a cut-off to identify individuals at high risk of ADHD.

Adult ADHD Self-Report Screening Scale for DSM-5

The Adult ADHD Self-Report Screening Scale for DSM-5 (ASRS-5; Ustun et al., 2017) assesses the adult-specific presentation of ADHD symptoms based on DSM-5 conceptualization (APA, 2013). It includes six items (e.g., “How often do you put things off until the last minute?”) in a 5-point Likert scale (0 = *never* to 4 = *very often*). As in the aBAARS-IV, the questions in this scale refer to behaviors that have occurred over the last 6 months. We used the Spanish version of the ASRS-5 that was administered in the original study (Coll-Martín et al., 2021). Cronbach’s alpha reliability scores in our final sample were .60, which is close to the .64 from our previous study. Ustun et al. established a threshold of 14 points in the ASRS-5 as the preferred bound for screening purposes.

ANTI-Vea

The ANTI-Vea (Luna et al., 2018) is a cognitive task that provides measures of the functioning of the three attentional networks (alerting, orienting and, executive attention), along

with measures of AV and EV (overall performance and decrement across time on task). The simultaneous assessment of these processes controls the order effect bias. The main characteristics of the task are depicted in Figure 2 and described in Sample Selection and Study Procedure section. We used the online version of the ANTI-Vea, which is freely available in multiple languages at https://anti-vea.ugr.es (Coll-Martín, Román-Caballero, et al., 2023). The ANTI-Vea has been psychometrically validated in lab and online settings with no substantial differences between the scores analysed in each task version (Luna et al., 2021).

In the present study, we focused on two types of ANTI-Vea measures. For AV, we assessed lapses, defined as trials with an excessively slow RT (i.e., RT > 600 ms) or no response to the red down counter. For EV, we measured hits, defined as trials in which the infrequent displacement of the central arrow is correctly detected. For each of the two measures, we considered both the overall task performance and the slope of the vigilance decrement across the six blocks of the task. This decrement manifests as an increase in the lapse rate and a decrease in the hit rate across blocks of trials. The internal consistency scores of the four indices were estimated using a permutation-based split-half approach (Parsons et al., 2019) with 10,000 random splits. The Spearman-Brown (SB) corrected coefficients in our final sample (see Table 2) were arguably close to the corresponding .96, .94, and .78 obtained in the original study (Coll-Martín et al., 2021) for lapse overall, hit overall, and lapse slope, respectively. The exception was the slope of hits, for which the reliability score in our sample (r_{SB} = .61) was substantially higher than in the original (r_{SB} = .27).

Data Analysis

As preregistered, data were preprocessed and analysed based on the original study (Coll-Martín et al., 2021). The entire workflow has been run and documented in three reproducible R

scripts (Version 4.2.1). We used the *tidyverse* collection of R packages (Wickham et al., 2019) for most of the data treatment and output visualization.

For the selection of ADHD symptom measures, we considered the period elapsed from the initial survey until the completion of the task. If this time interval was 90 days or less (44.2% participants of final sample), we used the questionnaires from the first survey. Conversely, if the period was longer than 90 days (53.8% participants of final sample), we used the ADHD self-reports they completed in the second phase of the study.[3] The score for each ADHD symptom scale (i.e., the cBAARS-IV, the aBAARS-IV, and the ASRS-5) was the sum of its items. Applying the procedure of the original study (Coll-Martín et al., 2021), the distribution of the cBAARS-IV and the aBAARS-IV scores were compared with a simulated normative sample.

Figure 1 shows the task sessions and participants excluded from the ANTI-Vea. Compared to the final sample, the level of ADHD symptoms among the participants excluded due to poor performance (i.e., more than 25% errors in ANTI trials) was negligibly lower for symptoms in childhood ($d_{\text{cBAARS-IV}} = –0.16$) and negligibly to slightly higher for adult symptoms ($d_{\text{aBAARS-IV}} = 0.03$; $d_{\text{ASRS-5}} = 0.22$). We calculated the percentage of lapses for AV stimuli and the percentage of hits for EV stimuli for each participant. Of the total number of lapses, 87.2% were due to excessively slow RT, while the remaining 12.8% were due to no response. For both lapse and hit percentages, vigilance decrement was calculated by estimating the linear slope of the outcome across the six blocks of the task. We used the *plyr* R package (Wickham, 2022) to compute the reliability of EV and AV measures.

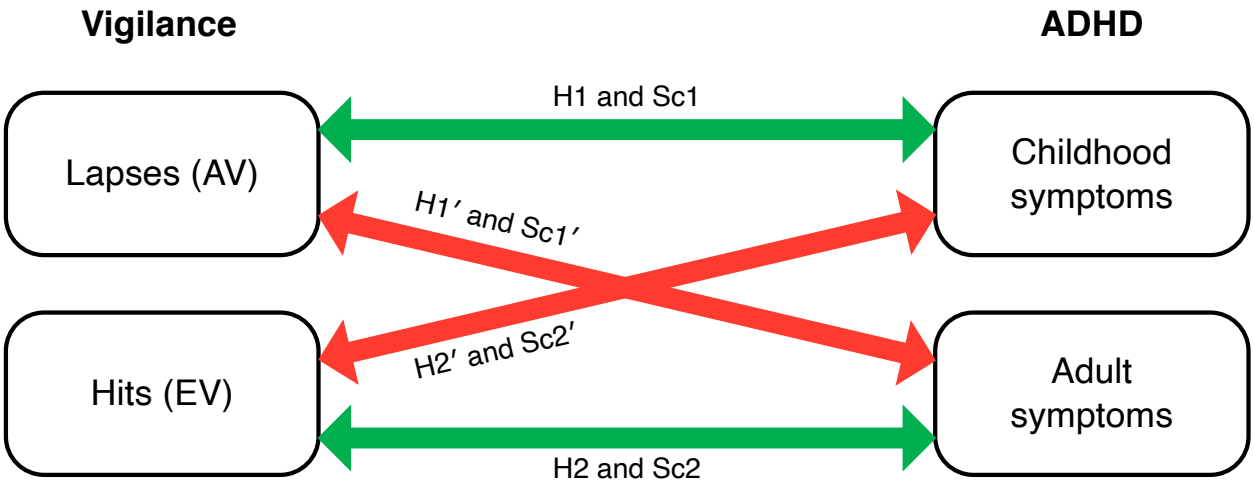
[3] There were six participants (2.0% of final sample) for whom more than 90 days elapsed from the initial survey to the task (*Mdn* = 144), but a second assessment of ADHD symptoms was not available. For these participants, the symptom measures collected in the baseline survey were used.

Regarding our preregistered statistical hypotheses, we distinguished between primary and secondary vigilance outcomes. Primary outcomes were the slopes of vigilance decrement across time on task, while secondary outcomes were the overall vigilance scores. Therefore, since the slope of lapses is positive while that of hits is negative, we hypothesised a positive correlation between the slope of lapses and childhood symptom severity measured with the cBAARS-IV (H1), as well as a negative correlation between the slope of hits and adult symptom severity measured with the ASRS-5 (H2). In the same vein, given that more lapses reflect poorer AV and fewer hits indicate poorer EV, we hypothesised a positive correlation between the overall percentage of lapses and the cBAARS-IV (Sc1), as well as a negative correlation between the overall percentage of hits and the ASRS-5 (Sc2). To focus more closely on the association with late-developing symptoms, H2 and Sc2 were partial correlations controlled for the cBAARS-IV. In parallel, we also tested H1′, H2′, Sc1′, and Sc2′, which represented the set of opposite statistical hypotheses, that is, those hypotheses relating each vigilance component to ADHD symptom severity in the age period opposite to that established in the neurodevelopmental model.[4] Figure 3 illustrates all the statistical hypotheses tested in our study.

[4] The neurodevelopmental model does not rule out the possibility that minimal impairment in executive functions could be related to ADHD symptoms in childhood. However, any such relationship should be substantially lower than nonexecutive dysfunctions (Halperin, 2016; Halperin & Schulz, 2006).

Figure 3

Statistical Hypotheses Tested in Our Study



Note. H1, H2, H1′, and H2′ are primary hypotheses measuring vigilance decrement, while Sc1, Sc2, Sc1′, and Sc2′ are secondary hypotheses measuring overall vigilance. All hypotheses are correlations between the two elements linked by the arrow. Correlations are positive in hypotheses involving lapses (AV) and negative in hypotheses involving hits (EV). Correlations involving symptoms in adulthood are controlled for symptoms in childhood. Green arrows represent statistical hypotheses derived from Halperin and Schulz's (2006) neurodevelopmental model, while red arrows are their opposite statistical hypotheses. AV = Arousal vigilance; EV = Executive vigilance; ADHD = Attention-deficit/hyperactivity disorder.

Since bivariate normality was violated for H1 and H2 (both *p*s < .001), we based on Kendall's rank-order correlation coefficients with the *correlation* R package (Makowski et al., 2022) to test the hypotheses. We conducted one-tailed contrasts at a significance level of α = .05 for primary statistical hypotheses (H1 and H2) and their opposite counterparts (H1′ and H2′), as well as α = .025 for secondary statistical hypotheses (Sc1 and Sc2) and counterparts (Sc1′ and

Sc2′) to control for multiple comparisons. Moreover, we based on the one-tailed limit of the 95% confidence interval (CI; corrected for multiple comparisons when applicable) to test whether the effect sizes we obtained were significantly lower, in absolute terms, than the effect sizes of interest[5] (for the effects of interest, see Table 1 and Supplemental Table 2). Additionally, we analysed the robustness of the results by performing a multiverse analysis (Steegen et al., 2016) for each statistical hypothesis. Finally, we explored the reach of the neurodevelopmental model to the ANTI-Vea core indices that measure arousal or executive processes.

Sample Size Justification

For the sample size justification, we followed Lakens's (2022) guidelines (see Supplemental Text 1 for a more detailed report prior to data analysis). The sample size collected in the first and second phases of the study (see Figure 1) was determined by the financial resources provided by our funders for participant payment. Our final sample consisted of 292 university students (49.0% women, 51.0% men; 18–30 years, $M = 21.7$, $SD = 2.7$; see Figure 4A). Most of the participants were Spanish (95.2%) and current university students (92.1%).[6] Five individuals (1.7%) reported a prior diagnosis of ADHD.

To appraise the informational value of the sample size for our preregistered hypotheses, we considered two types of effect size: the smallest effect size of interest (SESOI) and the expected effect size. As preregistered, we determined a true SESOI of $\rho = |.2|$ based on a combination of influential expert opinions (Willcutt et al., 2005), measurement error issues (Gignac & Szodorai, 2016), and recognition of heterogeneity in ADHD etiopathophysiology

[5] In statistical notation, this tests H_0: $|\rho| \geq$ |effect of interest| versus H_1: $|\rho| <$ |effect of interest|.

[6] Technically, 22 participants of the final sample (7.5%) were former university students. Additionally, due to an investigator error, one participant was neither pursuing nor had any previous university studies.

(Luo et al., 2019). According to recent considerations on measurement error in cognitive-behavioural tasks (Parsons et al., 2019), we based on the observed SESOI, which for correlations is obtained as follows:

$$r = \rho \sqrt{\text{reliability}(x) \times \text{reliability}(y)}, \quad (1)$$

where *x* and *y* are the two measures of the correlation. This formula provided the observed SESOIs as Pearson's correlation coefficients (*r*) for each of the four statistical hypotheses tested in our sample.

For the expected effect sizes, we based on the corresponding four Kendall's τ values from the original study (Coll-Martín et al., 2021). To counteract the potential overinflation of the effect sizes in the original study, we applied the Perugini et al.'s (2014) correction for replications. Following Gilpin's (1993) formula, we transformed Kendall's τ values into Pearson's *r* values. These Pearson's coefficients, along with the reliability of the measures in the original study, were used to estimate the true expected effect sizes through the Spearman's (1904) correction for attenuation formula—which is a rearrangement of Equation 1. Finally, each true expected effect was input into Equation 1 to obtain the *r* values that would be expected to be observed in our sample.

After estimating the two types of effects of interest, namely the SESOIs and the expected effects, we conducted simulations to compute the statistical power achieved to detect these effects across each of our four statistical hypotheses. Using the *faux* R package (DeBruine, 2021), we input the eight Pearson's coefficients and run 10,000 simulations to conduct the

corresponding hypotheses tests for Kendall's coefficients.[7] The minimal statistically detectable effect (i.e., the critical τ value) was also estimated for the primary and the secondary statistical hypotheses.

The results of the power analysis for the effect sizes of interest are shown in Table 1 (for the power analysis of the opposite statistical hypotheses, see Supplemental Table 2). With one exception, the statistical power to detect at least one of the two effects of interest was greater than 80% for each statistical hypothesis. Only for Sc2 the power achieved was at best slightly suboptimal ($1 - \beta = .69$). Furthermore, the SESOIs of all the hypotheses were higher than the critical τ value. Taking together, our sample size provides an arguably acceptable informational value with respect to the statistical hypotheses derived from the Halperin and Schulz's (2006) neurodevelopmental model.

[7] Although H2 and Sc2 are partial correlations controlled for the cBAARS-IV, they were simulated as zero-order correlations. This was to avoid that a null correlation between the cBAARS-IV and EV might overestimate the size of the partial correlation between the ASRS-5 and EV.

Table 1

Effect Sizes of Interest and Achieved Power to Detect Them Across Each Statistical Hypothesis

Hypothesis	Smallest effect of interest			Expected effect (from Coll-Martín et al., 2021)		
	r	τ	Power (1 – β)	*r*	τ	Power (1 – β)
H1	.16	**.10**	.82	.19	**.12**	.94
H2	–.12	**–.08**	.62	–.16	**–.10**	.83
Sc1	.19	**.12**	.87	.10	.06	.36
Sc2	–.15	**–.10**	.69	.11	.07	0

Note. $N = 292$. H1 and H2 are the primary statistical hypotheses, while Sc1 and Sc2 are the secondary statistical hypotheses. Kendall's τ values come from the Pearson's *r* values used to conduct the 10,000 simulations for power analysis (see Gilpin, 1993, for the formula to transform the correlation coefficients). Statistical power corresponds to one-tailed tests for Kendall's coefficients, with $\alpha = .05$ for primary statistical hypotheses and $\alpha = .025$ for secondary statistical hypotheses. Kendall's τ values above the minimal statistically detectable effect ($\tau = |.06|$ for primary hypotheses; $\tau = |.08|$ for secondary hypotheses) are in bold.

Results

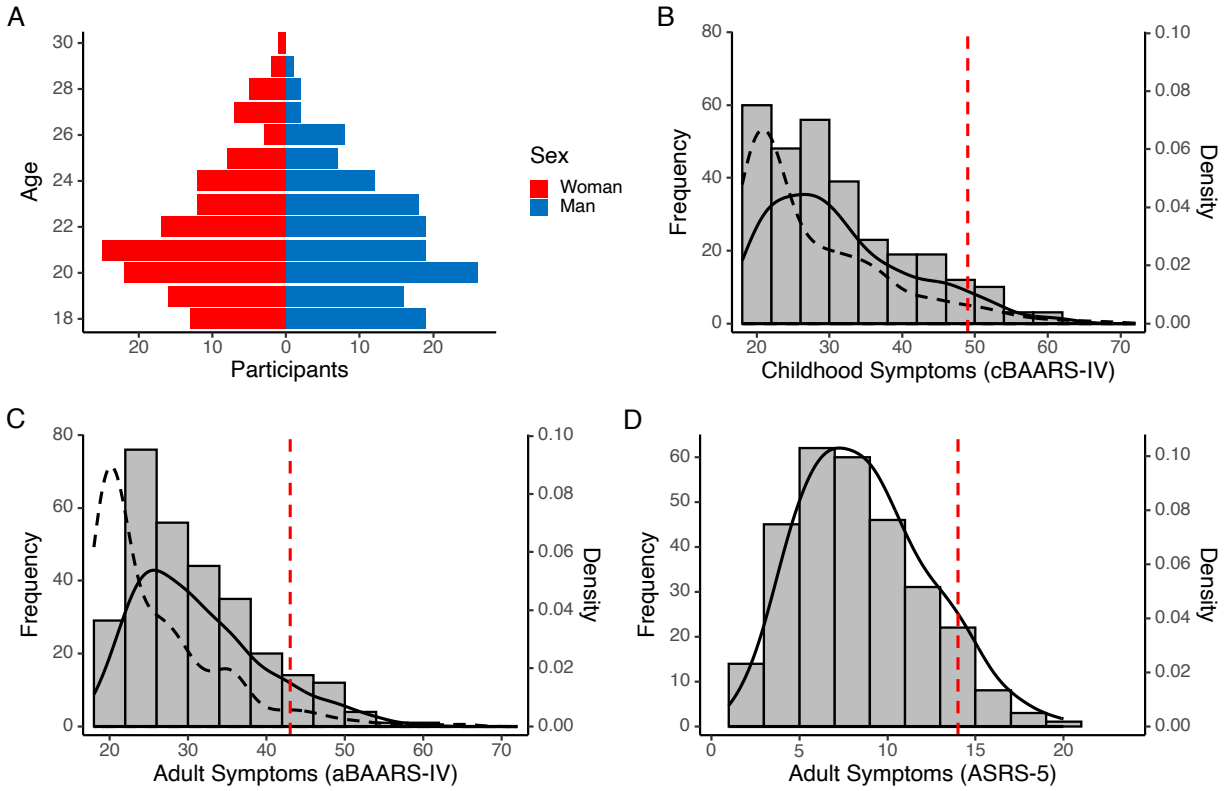
ADHD Self-Reports

Figure 4B–D shows the distribution of ADHD symptoms in the three scales, and Figure 4B–C also compares it to an estimated normative sample (for a detailed procedure and statistical report, see Supplemental Text 2). Taking together, ADHD symptoms in our sample of young university students was higher than in the general population of adults. Even so, the spread and

variability within each scale did not seem to differ from those observed in an estimated normative sample.

Figure 4

Basic Demographics and Distribution of Total ADHD Symptom Scores for Each of the Three Scales Compared to an Estimated Normative Sample



Note. $N = 292$. Panel A: Age-sex pyramid. Panels B–D: Histogram and black solid line represent the frequency and density curve of ADHD total scores in the study sample. The vertical dashed red line represents a threshold to identify individuals at risk of ADHD. Panels B and C: The dashed black line represents the density curve of ADHD total scores in an estimated normative sample. This normative and representative sample was obtained by extracting 292 quantiles from a large simulated sample ($N = 10{,}000$) from the percentile values available in (Barkley, 2011). ADHD = Attention-deficit/hyperactivity disorder; cBAARS-IV = Barkley Adult ADHD Rating Scale-IV: Childhood symptoms; aBAARS-IV = Barkley Adult ADHD Rating Scale-IV: Current symptoms; ASRS-5 = Adult ADHD Self-Report Screening Scale for DSM-5.

Unsurprisingly, the cBAARS-IV (M = 31.37, SD = 9.82), the aBAARS-IV (M = 31.16, SD = 8.14), and the ASRS-5 (M = 8.74, SD = 3.66) showed significant positive correlations among them, with effect sizes that are considered large in the field (Gignac & Szodorai, 2016). Concretely, for the cBAARS-IV with the aBAARS, $r(290) = .53$, $p < .001$, for the cBAARS-IV with the ASRS-5, $r(290) = .34$, $p < .001$, and for the aBAARS-IV with the ASRS-5, $r(290) = .75$, $p < .001$. As expected, the correlation between the two measures of symptoms in adulthood was higher than those between these measures and the one of symptoms in childhood (both ps < .001). Additionally, the correlation with childhood symptoms was stronger when the adult symptom items also corresponded to DSM criteria, as in the case of the aBAARS-IV, than when they were intended to reflect the adult-specific presentation of ADHD symptoms, as with the ASRS-5 ($z = 5.11$, $p < .001$).

Task Performance

Table 2 shows the means and standard deviations of the percentages of lapses and hits for both the vigilance decrement and the overall performance. In line with the original study (Coll-Martín et al., 2021), we found a positive linear slope of lapses, $t(291) = 7.18$, $d_z = 0.42$, and a negative linear slope of hits, $t(291) = –9.83$, $d_z = –0.58$, across time on task. Specifically, the lapses increased from 7.96% in Block 1 to 13.38% in Block 6, while the hits decreased from 78.94% in Block 1 to 68.90% in Block 6. Interestingly, the correlation between lapses and hits was not significant for either the overall scores ($r = –.09$, $p = .10$) or the decrement slopes ($r = –.11$, $p = .07$), revealing a dissociation between both vigilance components.

Preregistered Hypotheses

Table 2 shows the results of the preregistered hypotheses. For the primary outcomes (i.e., vigilance decrement), none of the statistical hypotheses—namely H1, H2, H1′, and H2′—were

statistically significant (all *p*s > .194). Even more so, each of these four correlation coefficients was significantly smaller in size than at least one of the two effects of interest (all *p*s < .05). Specifically, all correlation sizes, except that of H2, were lower than their corresponding SESOI. Moreover, the coefficients from H1 and H2 were smaller than their expected effect sizes.

Regarding the secondary outcomes (i.e., overall vigilance scores), the expected positive correlation between the cBAARS-IV and the percentage of lapses (Sc1) was only significant before correcting for multiple comparisons, $\tau = .07$, $p_{\text{uncorrected}} = .047$, $p_{\text{corrected}} = .094$. In addition, the hypothesised negative correlation between the ASRS-5 and the percentage of hits (Sc2) was not significant, $\tau = -.04$, $p_{\text{uncorrected}} = .128$. As expected, the correlation between the ASRS-5 and the percentage of lapses (Sc1′) was not significant, $\tau = -.00$, $p_{\text{uncorrected}} = .523$, and its effect size was smaller than the SESOI ($\tau_{\text{SESOI}} = .10$; $p_{\text{corrected}} < .05$). Surprisingly, the negative correlation between the cBAARS-IV and the percentage of hits (Sc2′) was significant, $\tau = -.09$, $p_{\text{corrected}} = .033$.

Table 2

Descriptive Statistics, Internal Consistency, and Kendall's Correlation Coefficients With ADHD Symptoms for Arousal Vigilance (AV) and Executive Vigilance (EV) According to Preregistered Hypotheses

Measure of vigilance	*M*	*SD*	r_{SB}	Correlation with ADHD symptom severity			
				Neurodevelopmental model		Opposite statistical hypotheses	
				Childhood ADHD	Adult ADHD	Childhood ADHD	Adult ADHD
Primary				H1	H2	H1′	H2′
% Lapse slope (AV)	1.09	2.60	.67	.00 [.07$^{<EI}$]			.01 [.07$^{<EI}$]
% Hit slope (EV)	–2.17	3.78	.61		–.03 [–.10$^{<EI}$]	.04 [–.03$^{<EI}$]	
Secondary				Sc1	Sc2	Sc1′	Sc2′
% Lapse overall (AV)	10.57	17.69	.98	.07† [.14]			–.00 [.07$^{<EI}$]
% Hit overall (EV)	73.08	18.28	.95		–.04 [–.12]	–.09* [–.16]	

Note. $N = 292$. Correlation with symptoms in childhood are zero-order correlations with the Barkley Adult ADHD Rating Scale-IV: Childhood symptoms (cBAARS-IV). Correlations with adult symptoms are partial correlations with the Adult ADHD Self-Report Screening Scale for DSM-5 (ASRS-5) after controlling for the cBAARS-IV. Correlational tests are positive for AV and negative for EV. Neurodevelopmental model encompasses the preregistered hypotheses derived from the Halperin and Schulz's (2006) model.

Opposite statistical hypotheses are included for comparison purposes. One-tailed, 95% (primary outcomes) or 97.5% (secondary outcomes) limits of confidence intervals are in brackets. ADHD = Attention-deficit/hyperactivity disorder; r_{SB} = Spearman-Brown split-half reliability coefficient.

$^{\dagger}p_{uncorrected} < .05$, one-tailed. $^{*}p_{corrected} < .05$, one-tailed. $^{<EI}p_{corrected} < .05$, one-tailed inferiority test: the limit of the confidence interval excludes at least one of the two effects of interest (EIs; for the EIs, see Table 1 and Supplemental Table 2).

Multiverse Analyses

To generate the multiverse of analyses, we identified five decision points in the analytical process with more than one apparently reasonable choice, the first being the preregistered option: participants with more than 90 days from survey to task (n = 6; retained vs. excluded), poor task performers (n = 14; excluded vs. retained), task threshold to compute the vigilance indices (absolute vs. relative), correlation coefficient (Kendall vs. Spearman vs. Pearson) and type of correlation (zero-order vs. partial[8]). Furthermore, we incorporated the aBAARS-IV as a secondary measure of symptoms in adulthood (preregistered as Ss2). This provided 48 valid specifications to analyse the correlation between each of the four vigilance indices and each of the three ADHD measures (i.e., 576 total estimates). The distribution of the estimates within the four multiverses is illustrated in Figure 5 (for a more comprehensive report of the method and results of these analyses, see Supplemental Table 4).

Figure 5A–B shows that the lack of significant findings in the preregistered H1 and H2 was robust across the different analytical options, including the use of a secondary measure of ADHD symptoms in adulthood (i.e., the aBAARS-IV). Regarding their opposite counterparts, we found that 20 out of the 96 analytical scenarios in H1′ (20.8%) were statistically significant. A closer inspection revealed that most of these significant findings (90.0%) computed the lapse index based on a relative threshold (i.e., trials with an RT higher than the participant's mean plus one standard deviation or trials with no response). This type of lapse index computation exhibited very low reliability scores, albeit probably biased upwards when poor performers were retained for the analyses (see Supplemental Figure 1). On the contrary, for the analytical

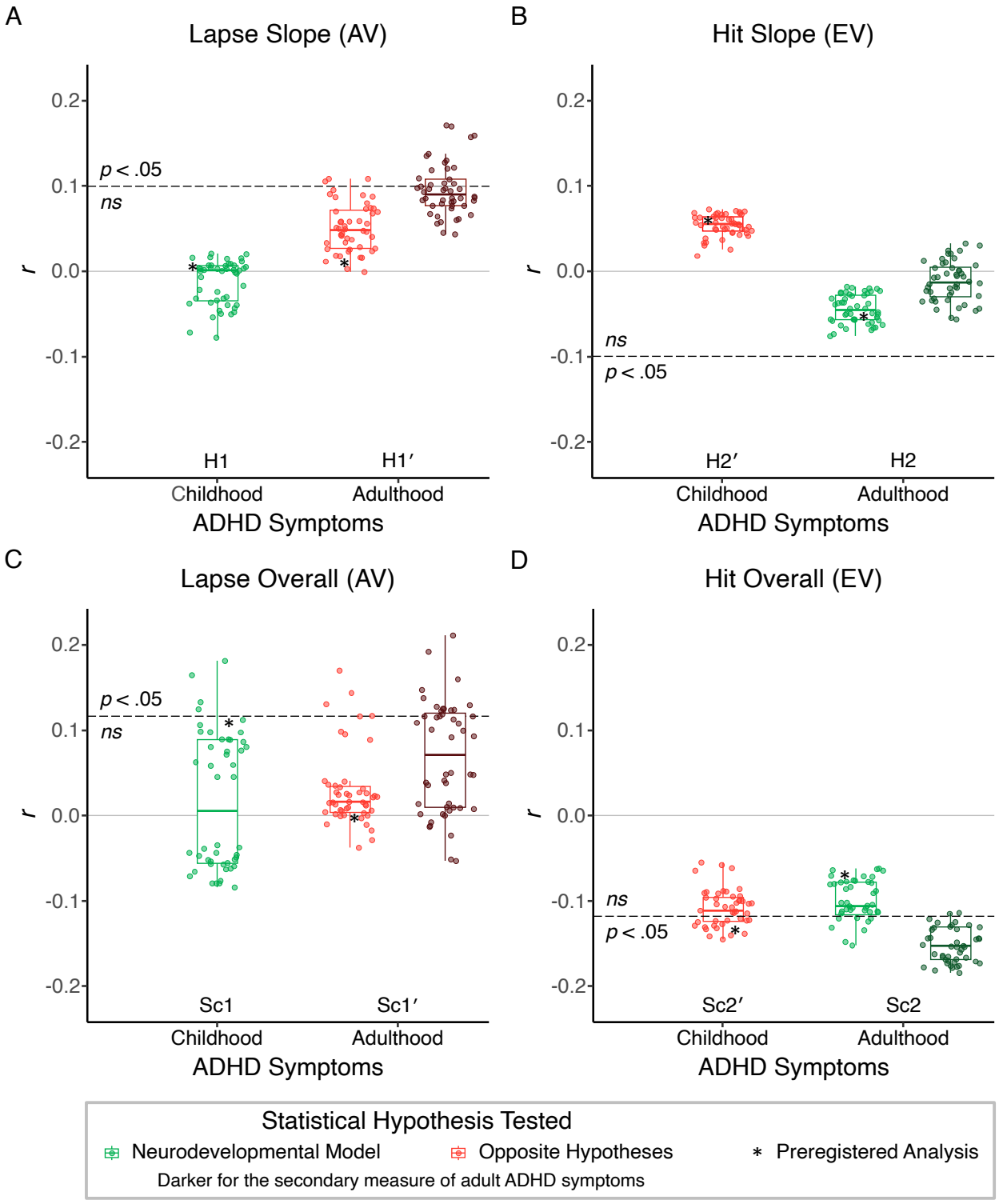
[8] For the ASRS-5, partial correlation was the preregistered option.

combinations that employed an absolute threshold to compute lapses, the rate of significant findings was below chance (2.1%). As for H2′, the null results were consistent across all analytical choices.

In contrast to the primary hypotheses, secondary hypotheses (Figure 5C–D) showed a rather mixed pattern of statistical significances across analytical scenarios. Sc1 yielded 8.3% significant contrasts, which doubled to 16.7% when only analyses with an absolute threshold for lapses were considered. Regarding Sc2, the percentage of statistically significant scenarios dramatically varied from 16.7% in the ASRS-5 to 95.8% in the aBAARS-IV. This difference between adult measures of ADHD symptoms was also manifested in Sc1′, where 20.8% and 54.2% analytical combinations were significant for the ASRS-5 and the aBAARS-IV, respectively. Sc2′, the only statistical hypothesis whose preregistered contrast was significant, yielded 35.4% scenarios with positive findings. Since most of our analytical options do not follow the linear regression model, the robustness of the secondary hypotheses could not be inferentially evaluated with available approaches (e.g., Simonsohn et al., 2020).

Figure 5

Distribution and Box Plot of Correlation Coefficients of Arousal Vigilance (AV) and Executive Vigilance (EV) With ADHD Symptoms Across the Multiverse of Reasonable Analytical Options



Note. $N = 292$. Kendall and Spearman correlation coefficients were transformed into Pearson's *r* values to allow for comparison of effects (formulas based on Gilpin, 1993). ADHD symptoms in childhood and adulthood are measured with the Barkley Adult ADHD Rating Scale-IV: Childhood symptoms (cBAARS-IV) and the Adult ADHD Self-Report Screening Scale for DSM-5 (ASRS-5), respectively. The secondary measure of ADHD symptoms in adulthood refers to the Barkley Adult ADHD Rating Scale-IV: Current symptoms (aBAARS-IV). The horizontal dotted grey lines represent the significance threshold for the correlations. This significance threshold is based on one-tailed tests (positive for AV and negative for EV), corrected for multiple comparisons in secondary statistical hypotheses (i.e., Panels C and D). Panel D: Due to the variety of statistical approaches, one statistically significant coefficient in the ASRS-5 ($r = –.117$) and two in the aBAARS-IV (rs = –.116 and –.117) appear above the significance threshold line, while one nonsignificant coefficient in the ASRS-5 ($r = –.120$) appear below such line. ADHD = Attention-deficit/hyperactivity disorder.

Exploratory Analyses of all Arousal and Executive Task Indices

To analyse the relationships between ADHD symptoms and both arousal and executive processes more broadly, we computed all ANTI-Vea indices of these neurocognitive domains (see Supplemental Table 5 for descriptive statistics and correlations among them). They included eight measures of arousal: lapses in AV trials (slope and overall), mean RT in AV trials (slope and overall), standard deviation of the RT in AV trials (slope and overall), and the alerting index (i.e., no tone minus tone conditions) in ANTI trials (RT and percentage of errors). They also included six executive measures: hits in EV trials (slope and overall), false alarms in EV trials

(slope and overall), and the congruency index (i.e., incongruent minus congruent conditions) in ANTI trials (RT and percentage of errors).

The set of 14 ANTI-Vea indices was used to predict each of the three measures of ADHD symptoms (i.e., the cBAARS-IV, the ASRS-5, and the aBAARS-IV) through multiple linear regression models. To select the predictors of each model, we employed a bidirectional stepwise regression method aimed at minimising the Akaike information criterion (AIC). The results (Table 3) showed that three, six, and five ANTI-Vea indices were selected to predict the cBAARS-IV, the ASRS-5, and the aBAARS-IV, respectively. The measure of symptoms in childhood was also entered and selected in both models of adult symptoms. Crucially, both arousal and executive task indices uniquely predicted ADHD symptoms across the three models.

Table 3

Multiple Regressions of Three Models of ADHD Symptoms as a Function of Arousal and Executive Task Indices

Predictor	Model 1: Childhood (cBAARS-IV)		Model 2: Adulthood (ASRS-5)		Model 3: Adulthood (aBAARS-IV)	
	B (*SE*)	β	*B* (*SE*)	β	*B* (*SE*)	β
Arousal						
Lapse overall	0.10 (0.03)	.18**	0.05 (0.02)	.23*		
RT *M* overall			–0.01 (0.00)	–.18		
RT *SD* overall					0.04 (0.01)	.15**
Alerting RT			0.01 (0.01)	.08	0.03 (0.01)	.13**
Executive						
Hit overall	–0.07 (0.03)	–.12*	–0.02 (0.01)	–.11*		
FA slope	0.89 (0.33)	.15**			–0.49 (0.23)	–.10
FA overall			0.12 (0.04)	.17**		
Congruency RT			0.01 (0.01)	.11*	0.03 (0.02)	.11*
Congruency errors					0.15 (0.10)	.07
R^2	.08					
ΔR^2			.07		.07	

Note. n = 289 (three participants from the final sample were dropped due to an incorrect task setting for FAs). The predictors of each model were selected from a set of 14 relevant task indices (eight of arousal and six executive) through a bidirectional stepwise regression method based on the Akaike information criterion (AIC). The two models predicting adult symptoms

also include symptoms in childhood (i.e., the cBAARS-IV) as a predictor variable. As such, the incremental variance (ΔR^2) of these models indicates the proportion of explained variance above and beyond that accounted for by a model only including childhood symptoms. ADHD = Attention-deficit/hyperactivity disorder; cBAARS-IV = Barkley Adult ADHD Rating Scale-IV: Childhood symptoms; ASRS-5 = Adult ADHD Self-Report Screening Scale for DSM-5; aBAARS-IV = Barkley Adult ADHD Rating Scale-IV: Current symptoms; RT = Reaction time; FA = False alarm.

$^{*}p < .05$, one-tailed (left-tailed for hit overall and right-tailed for the rest of predictors, as per Coll-Martín et al., 2021). $^{**}p < .01$, one-tailed.

Discussion

Are alterations in AV associated with higher ADHD symptoms in childhood (H1), while deficits in EV are related to late-developing ADHD symptoms in adulthood (H2)? This dissociation, predicted by the Halperin and Schulz's (2006) neurodevelopmental model, was supported in Coll-Martín et al. (2021). Here we designed a preregistered, closed replication of that study specifically aimed at testing the model. Based on a dimensional framework of ADHD, we assessed retrospective and current self-reported symptoms in a community sample of university students who performed an online cognitive task: the ANTI-Vea. The final sample size ($N = 292$) allowed our preregistered hypotheses to achieve an arguably acceptable statistical power for the effects of interest. Despite this, our main results failed to replicate the previous study: Indeed, only the unexpected (i.e., opposite to the neurodevelopmental model) negative correlation between EV and symptoms in childhood was significant. Although multiverse and

exploratory analyses yielded some significant findings, they neither supported the dissociation pattern proposed by the neurodevelopmental model.

Our unsuccessful replication of Coll-Martín et al. (2021) was rather clear and consistent. Not only were our correlation coefficients for H1 and H2 not significantly different from zero, but they also were significantly smaller in size than an attenuated estimation of the effect sizes in the original study. Even testing H2 as a simple correlation, as conducted in the original study, was not significant. The minimal conceptual differences between the original study and this replication—namely, the task setting (lab vs. online) and the composition of the university sample (mostly women from the same degree subject vs. sex-balanced sample from the entire student community)—were unlikely to account for the huge discrepancies in the results.[9] Instead, it is far more plausible that the findings of H1 and H2 in the original study were false positives, given its exploratory approach and lower control of the Type I error. This highlights the importance of close replications with adequate control of both types of error rates to improve the reliability of a given literature.

In contrast to Coll-Martín et al. (2021), our study failed to support the neurodevelopmental model. Notably, unlike most previous empirical research, our study selected a clear and valid nonexecutive measure to test the first prediction of the model. Specifically, the AV indices from the ANTI-Vea are thought to reflect the noradrenergic mechanisms of the hindbrain mediating arousal (Luna et al., 2018). According to the neurodevelopmental model, alterations in this subcortical system are a potential cause of ADHD and remain stable

[9] We tested H1 and H2 in our subsample of women. None of the Kendall's coefficients were greater in size than the minimal statistically detectable effect in the full final sample. The same was true when H2 was tested as a simple correlation, as in the original study.

throughout the lifetime. The conceptual and empirical dissociation of AV from EV in the ANTI-Vea—as evidenced by the low correlation found in our data—support the task as a useful tool in the neurocognitive research on ADHD.

Another measurement issue, in this case related to the second prediction of the neurodevelopmental model, was the use of two conceptually different scales of ADHD symptoms in adulthood. Our primary measure was the ASRS-5, which assess the adult presentation of ADHD, including adult-specific items (e.g., frequency of difficulties in unwinding and relaxing when having time to oneself). In addition, we administered the aBAARS-IV as a secondary measure, which consists of traditional DSM items with identical behaviour and wording (e.g., frequency of fidgeting with hands or feet or squirming in seat) as its childhood counterpart (i.e., the cBAARS-IV). While the debate between using adult-specific versus DSM-based items for ADHD diagnosis has been addressed elsewhere (e.g., Sibley et al., 2012), differences in sensitivity of these measures to neurocognitive variables has not been formally studied. Although descriptively, our multiverse and exploratory analyses suggest that DMS-based items could be somewhat more sensitive to attentional functioning than adult-specific items, especially considering that the former measure is potentially more affected when controlling for childhood symptoms. Alternatively, the higher reliability and skewness of the aBAARS-IV compared to the ASRS-5 could explain such differences in the size of the correlations.

Is impaired vigilance decrement core to ADHD?

We used measures of vigilance decrement for our primary hypotheses (H1 and H2) and their opposite counterparts (H1′ and H2′). Compared to overall vigilance scores, the decline in performance over time has been less studied in ADHD, despite being considered by some as the

defining feature of sustained attention (Huang-Pollock et al., 2012; Tucha et al., 2017). Surprisingly, our four correlation coefficients were significantly smaller in size than at least one of their corresponding effects of interest. Furthermore, the lack of a correlation statistically different from zero in the expected direction was robust across the sets of reasonable analytical options, as shown in the multiverse analyses. Only one exploratory analysis in our stepwise multiple linear regression, namely the positive association between the rate of false alarms and childhood symptoms, was significant.

Considering that our analyses had an acceptable statistical power, our lack of significant results for the relationship between vigilance decrement and ADHD symptoms was theoretically unexpected. However, they are in line with other clinical and community studies that used moderate to large samples (i.e., $N > 150$; Aduen et al., 2020; Huang-Pollock et al., 2020; Huang-Pollock et al., 2012). In contrast, one study comparing ADHD children with controls ($n > 200$ per group) found that the former had a higher vigilance decrement in response speed and consistency (each $d_s > 0.3$; Weyandt et al., 2017). While this discrepancy needs to be solved, one possibility might be that vigilance decrement is not substantially related to ADHD symptomatology. On the contrary, momentary attentional fluctuations, the other component of vigilance (Esterman & Rothlein, 2019), could be behind the impaired sustained attention in ADHD. In our study, preliminary support for this idea comes from the association between a higher rate of lapses and higher ADHD symptom severity in 30.1% of the multiverse scenarios with an absolute lapse threshold as well as in the exploratory multiple regression models.

Is the neurodevelopmental model a useful theory?

Looking at the big picture, the most consistent pattern across our results was the failure to support the neurodevelopmental dissociation proposed by the Halperin and Schulz's (2006)

model. In our preregistered analyses, we found that the rate of hits (EV) negatively correlated with symptoms in childhood but not in adulthood, which is the opposite to the theoretical prediction. Furthermore, our exploratory regression analyses were far from suggesting any dissociation: Both arousal and executive processes independently predicted ADHD symptoms in childhood and adulthood. Although contrary to Coll-Martín et al. (2021), our lack of a dissociation supporting the model is in line with several studies (Coghill et al., 2014; Coll-Martín, Sonuga-Barke, et al., 2023; Gmehlin et al., 2016; McAuley et al., 2014; van Lieshout et al., 2013).

Going further, some of the core neurocognitive phenomena for which the neurodevelopmental model was formulated are arguably better explained by alternative accounts. For example, the neurodevelopmental model overemphasises the prefrontal cortex subserving executive functions as the only area responsible for changes in ADHD symptoms across the lifespan. This contrasts with evidence suggesting that executive functions are implemented in distinct neural networks involving multiple brain regions (e.g., Dosenbach et al., 2008). Furthermore, the neurodevelopmental model proposal of brain lesions or insults as causes or developmental moderators of ADHD is inconsistent with the current neurodiversity model, which considers ADHD as the extreme expression of a temperamental trait (Sonuga-Barke & Kostyrka-Allchorne, 2022; Sonuga-Barke & Thapar, 2021). The latter model can provide a more straightforward explanation for why brain stimulation through modern neurotherapeutics or computer-based cognitive training has at best limited effects on core ADHD symptoms (Rubia, 2022; Westwood et al., 2023). In addition, the heterogeneity of neurocognitive alterations associated to ADHD symptoms within each stage of development and its change trajectories is

better accounted for by models of multiple developmental pathways (Nigg et al., 2005; Sonuga-Barke, 2005).

Taking together, it can be stated that the Halperin and Schulz's (2006) neurodevelopmental model in its current form has substantially lower explanatory capacity than other alternative theories in the field. More broadly, assuming that there is no developmental dissociation in the neurocognitive processes underlying ADHD symptoms has important implications. If late-developing symptoms are pathophysiologically similar to symptoms in childhood, then late-onset ADHD cases—or a substantial proportion of them—would be close in nature to child-onset ADHD. Therefore, translational interventions for ADHD symptoms should be designed to target the same underlying neurocognitive processes regardless of the age of individuals and impairment onset.

Limitations

We have identified three main limitations in the design of our study. The first regards the generalization of our findings. Although our community sample was sex balanced, it was mainly made up of university students, which is not fully representative of the young or general adult population. Indeed, ADHD symptoms in childhood have been associated with lower educational attainment in adulthood (Galéra et al., 2012; Pingault et al., 2011). Despite this sampling bias, our statistical analyses did not suggest that ADHD symptom scores were lower or more homogeneous in our sample than in a representative community sample. In any case, replication of our findings with a more representative sample is warranted. Additionally, although the dimensional framework assumes that the neurocognitive correlates of ADHD symptoms remain constant throughout a continuous trait, the empirical extension of our results to clinical populations is crucial.

Second, our assessment of ADHD symptoms in childhood and adulthood relied solely on self-reports, with the measurement of childhood symptoms being retrospective. Although self-reported measures of ADHD symptoms may capture a broader dimension, they seem to be less sensitive than parent-reports to some neurocognitive outcomes (Du Rietz et al., 2016; Riglin et al., 2022). In addition, the validity of retrospective reports of ADHD symptoms has been questioned due to potential recall bias, with longitudinal studies finding a modest correlation between prospective and retrospective parent ratings of symptoms in childhood (e.g., $r = .39$; von Wirth et al., 2021). However, given the fluctuating trajectory of ADHD symptoms throughout development (Sibley et al., 2022; Stern et al., 2020), part of the mismatch between both measures may be better explained by differences in the time span assessed by each symptom scale (e.g., last 6 months vs. whole childhood) rather than recall bias. Of note, Lundervold et al. (2021) found a 7-year test–retest reliability score of .89 for a retrospective self-report measure of ADHD symptoms in childhood. Looking on the bright side, our study design held constant the rater and the time of assessment, thereby controlling for biases related to these factors. In any case, the integration of distinct but complementary assessment methods is fundamental to advance the understanding of the neurocognitive processed associated to ADHD symptoms across lifespan.

The third limitation is about the statistical conclusion validity of our preregistered hypotheses. Our sample size was relatively large in this literature. Even so, it was far from achieving a reasonable statistical power to test for differences between correlations, the most pertinent analysis for contrasting the hypotheses of the neurodevelopmental model in our design. In fact, we would have needed around twice our sample size to perform such a test. Descriptively, none of the differences in the correlation coefficients between the ADHD measures of each vigilance outcome (e.g., τ in H1 minius τ in H1′; see Table 2) is higher in size

than any SESOI (see Table 1 and Supplemental Table 2). Despite this constraint that prevented us from comparing correlations inferentially, our study employed a thoughtful power analysis for single nonparametric correlations, accounting for measurement error to estimate the effects of interest. This approach not only allowed for transparent and accurate reporting of statistical power for each preregistered statistical hypothesis, but also enabled that all our primary hypotheses were statistically conclusive—by comparing each CI limit with the effects of interest. Although these practices are essential to evaluate the informational value of an empirical study, they are rarely implemented (Lakens, 2022; Parsons et al., 2019).

Conclusion

Assuming a dimensional framework, our unsuccessful preregistered close replication of Coll-Martín et al. (2021) did not support the Halperin and Schulz's (2006) neurodevelopmental model of ADHD: Only one unpredicted correlation was significant. Neither our exploratory findings were in line with the developmental dissociation hypothesised by the model: Both arousal and executive task indices uniquely predicted both ADHD symptoms in childhood and adulthood. Based on our findings, translational interventions for ADHD symptoms should target the same underlying attentional deficits regardless of the age of individuals and onset of the impairment. Future studies should include complementary assessment methods of ADHD symptoms, clinical groups, and other neurocognitive domains to qualify and extend our tentative recommendations.

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Supplemental Materials for

Attention-Deficit/Hyperactivity Disorder Symptoms as Function of Arousal and Executive Vigilance: Testing the Halperin and Schulz's Neurodevelopmental Model in an Adult Community Sample

Tao Coll-Martín, Hugo Carretero-Dios, and Juan Lupiáñez

Supplemental materials other than document file format are openly available at https://osf.io/vqgms/

Supplemental Text 1

Sample Size Justification Report Prior to Conduct the Data Analysis

The following text has been obtained through the Shiny app that accompanies Lakens' (2022) article titled "Sample Size Justification". Note that the output was generated on June 17th, 2022, prior to data analysis. At that time, we did not know the reliability scores of the measures in our sample, and thus the power calculation was based on the values obtained in the original study. Furthermore, we did not know that the normality assumption had been violated, and therefore, the power calculation was based on Pearson's correlation coefficients instead of Kendall's rank-order correlation coefficients, as we ultimately did.

Below are four sections of the sample size justification. Part A contains a description of the population, as well as a description of the resource constraints that determine how much of the population can be sampled. In Part B a description of which effect sizes are of interest is provided. In Part C an overview of the inferential goal of the study is specified. In Part D the sample size that will be collected is reported, and the informational value of the study is evaluated.

A: Sample Description

Description of the population.

```
We have recruited 318 university students (50% women). They were initially re
cruited through advertisements on the distribution list of our university. Co
ncretely, a total of 2003 respondents completed an online survey in exchange
for the opportunity to win two prizes of 200 € in a raffle. We excluded parti
cipants with invalid responses (i.e., extreme response times or careless resp
onses), serious diseases, psychiatric disorders or conditions other than ADH
D, age over 35, no current or former university studies, or repeated response
s. After exclusions, we invited participants to collaborate in the second pha
se of the research consisting in performing an online cognitive task. Invitat
ions were sent out by email until nearly 300 responses were collected. They r
eceived 6 € as compensation for their voluntary participation. We stratified
for sex to have equal representation in our final sample. The population we w
ant to generalise our findings consists of general young adults.
```

Can you collect data from the entire population?

```
no
```

Description of resource constraints.

```
The maximum sample size we could collect (N ≈ 300) was determined by the pecu
niary amount the funder could provide us for participants payment.
```

B: Effects of Interest

Information about the Smallest Effect Size of Interest. The smallest effect size of interest size is specified as a correlation of 0.2. The following details were provided about the smallest effect size of interest:

```
For our planned hypothesis tests H1 and H2 we specify the smallest effect siz
e of interest (SESOI). In the context of ADHD theories based on neurocognitiv
e processes, an influential meta-analysis on executive functions in ADHD cons
idered a moderate difference (ds ≈ 0.5; equivalent to r = 0.24; and this, in
turn, equivalent to a true correlation (ρ) = .32, Gignac & Szodorai, 2016) as
insufficient to validate the proposed mechanism as central in the disorder (W
illcutt et al., 2005). Given the heterogeneity in ADHD etiopathophysiology, w
e have set a less exigent SESOI of ρ = .2. Below this threshold, the theory w
ould be too imprecise, and probably other neurocognitive predictors should be
added to reframe the theory and make it relevant.


Notably, based on current considerations in cognitive-behavioural measurement
s (Parsons et al., 2019), we will focus on the observed SESOI, which for corr
elations would be obtained as follows:


Observed SESOI (r) = True SESOI (ρ) * √(Reliability (x) * Reliability (y).


Although the final observed SESOI will be computed from the reliability of th
e measures in the present study, we provide an estimation from our previous s
tudy using a virtually identical task (Coll-Martín et al., 2021), where: SESO
I would be r = .167 in H1 and r = −.083 in H2 (for the zero-order correlatio
n). Finally, in the likely event that the bivariate normality assumption is v
iolated for either H1 or H2, we will use Kendall's τ coefficient as a measure
of both effect sizes. The equivalent coefficient will be obtained from Gilpin
(1993).
```

Information about the Minimal Statistically Detectable Effect. The minimal statistically detectable effect is specified as a correlation of 0.10. The following details were provided about the minimal statistically detectable effect:

```
In our previous study we had 5.83% of invalid participants. To be conservativ
e, and given that the present study is online, we plan for 10% of invalid par
ticipants, so 318-32 = 286 valid responses. In G*Power (Version 3.1.9.6) we c
ompute the critical r-value:


Exact - Correlation: Bivariate normal model

Options: exact distribution

Analysis: Sensitivity: Compute required effect size

Input: Tail(s) = One

Effect direction = r ≥ ρ

α err prob = 0.05

Power (1-β err prob) = 0.8

Total sample size = 286

Correlation ρ H0 = 0

Output: Lower critical r = 0.0974574

Upper critical r = 0.0974574

Correlation ρ H1 = 0.1464436
```

Information about the Expected Effect Size. The expected effect size is based on a previous study. The expected effect size is a Correlation of NA [since there is more than one expected effect, no specific value appears]. An evaluation of the similarity of the previous study with the planned study, a citation of the previous study, and details about the effect size from the previous study are provided below:

```
We based on our previous study (Coll-Martín et al., 2021) to estimate the exp
ected effect size for the present study. Indeed, this planned study can reaso
nably be considered a close replication. Both studies recruited participants
from the same university, although the original study was more restricted to
students of specific degrees. Similarly, both studies employed the same type
of questionnaires for ADHD symptoms and a virtually identical task to measure
vigilance, being the main difference in the setting where the task was perfor
med: laboratory (original) or virtual (present replication). Finally, for bot
h studies the procedure consisted of a first phase where the questionnaires w
```

```
ere answered, followed by a second phase where the cognitive task was carried
out.
```

An evaluation of the uncertainty in the effect size estimate in the previous study, and how this is dealt with (e.g., choosing a more consevative estimate) is provided below:

```
In the original study, the effect size was r = .215, 95% CI [.032, .385] for
H1 and r = −.225 95% CI [−.379, −.023] for H2. The next section will attempt
to deal with uncertainty, along with publication bias.
```

An evaluation of whether the effect size estimate of the previous is unbiased, and if not, any approach to correct for bias, or decisions about the use of a more conservative effect size estimate is provided below:

```
The effects sizes of the original study are likely to be inflated, as they we
re the main significant findings that could lead to the publication of the ma
nuscript. Applying the suggestion of Perugini et al. (2014) for replication s
tudies, a better estimation of the expected effect size is the closer-to-zero
-bound of the 60% confidence interval in the original study. Therefore, the e
xpected effect size for the present study would be r = 137 for H1 and r = −.1
32 for H2.
```

The following information about the Sensitivity Power Analysis has been provided:

```
A sensitivity analyses with our expected final sample size (N = 286) indicate
s that, for α = .05, a power of 80% correspond to an r-value = |.146| for one
-tailed tests.


Exact - Correlation: Bivariate normal model


Options:    exact distribution


Analysis:   Sensitivity: Compute required effect size
Input:      Tail(s)                     =   One
            Effect direction         =  r ≤ ρ
            α err prob                  =   0.05
            Power (1-β err prob)        =   0.8
            Total sample size           =   286
            Correlation ρ H0            =   0
```

```
Output: Lower critical r            =   -0.0974574
            Upper critical r            =   -0.0974574
            Correlation ρ H1            =   -0.1464436


This effect size for which we have sufficient power to detect is lower than t
he SESOI in H1 (r = .167), slightly higher than our expected effect sizes in
H1 (r = .137) in H2 (r = −.132), and far higher than the SESOI in H2 (r = −.0
83; for zero-order correlation).
```

C: Inferential Goal

The following information about the inferential goal related to statistical power has been provided.

The inferential goal is to perform a hypothesis test with a certain statistical power, computed by a sensitivity power analysis. The chosen alpha level is 0.05 A justification for the chosen alpha level and desired power (or for a sensitivity power analysis, the achieved power for effects of interest), and details of the power calculation (preferably in reproducible code) is provided below:

```
We have collected a total of 318 participants. In the previous study, we had
a 5.83% of invalid responses (due to poor task accuracy) for the main analyse
s. Here we apply a conservative estimation of a 10% of invalid responses, lea
ving a total of (318 − 32) 286 participants with valid answers. Given this fi
nal sample size, and considering a critical α = .05, we achieve the following
statistical power for the effects of interest in one-tailed correlations:

-For the SESOI, our power is 88.43% for H1 (r = .167) and 40.40% for H2 (r =
−.083; for zero-order correlation).

-For our expected effect size, our power is 75.15% for H1 (r = .137) and 72.3
5% for H2 (r = −.132).
```

D: Informational Value of the Study

Based on the resource constraints, the effects of interest, and the inferential goals, the following evaluation of the informational value of the study has been provided.

Given the following resource constraints:

```
The maximum sample size we could collect (N ≈ 300) was determined by the pecu
niary amount the funder could provide us for participants payment.
```

and given a smallest effect size of interest size of Correlation = 0.2, a minimal statistically detectable effect of Correlation = 0.10, an expected effect size of Correlation = NA, an evaluation of effects one has sufficient power to detect based on a sensitivity power analysis as described below:

```
A sensitivity analyses with our expected final sample size (N = 286) indicate
s that, for α = .05, a power of 80% correspond to an r-value = |.146| for one
-tailed tests.


Exact - Correlation: Bivariate normal model


Options:    exact distribution


Analysis:   Sensitivity: Compute required effect size
Input:      Tail(s)                     =   One
            Effect direction         =  r ≤ ρ
            α err prob                  =   0.05
            Power (1-β err prob)        =   0.8
            Total sample size           =   286
            Correlation ρ H0            =   0
Output: Lower critical r            =   -0.0974574
            Upper critical r            =   -0.0974574
            Correlation ρ H1            =   -0.1464436


This effect size for which we have sufficient power to detect is lower than t
he SESOI in H1 (r = .167), slightly higher than our expected effect sizes in
H1 (r = .137) in H2 (r = −.132), and far higher than the SESOI in H2 (r = −.0
83; for zero-order correlation).
```

and given the inferential goal based on a sensitivity power analysis with an alpha level of 0.05, the sample size in the planned study will consist a total of 286 participants, each contributing 1 observations. The following additional details about the sample size were provided:

```
The estimation of the final sample size (N = 286; after exclusion of particip
ants with poor accuracy) from the initial sample size of 328 participants is
based on a protective expectation of 10% invalid responses (almost twice as m
uch as in the original study).


Moreover, based on current considerations in cognitive-behavioural measuremen
ts, our effect of interest considers measurement error as per the following f
ormula:


Observed SESOI (r) = True SESOI (ρ) * √(Reliability (x) * Reliability (y)).
```

An explanation of the informational value of the sample size that will be collected, given any resource constraints, the effects of interest, and the inferential goal, is provided below:

```
With an estimated final sample of 286 participants, we expect to reach a dece
nt power to test two relevant predictions derived from the Halperin and Schul
z's (2006) neurodevelopmental model. In the case of H1, we are likely to have
sufficient power for the SESOI (88.43%), which is higher than the expected ef
fect size. As per H2, although our study is rather underpowered for the SESOI
(40.40%), the power is only slightly suboptimal for the expect effect size (7
2.35%). Considering that this latter estimation has followed a conservative s
trategy (called safeguard power analysis), the achieved power can be deemed a
s acceptable. Taking together, we think that our sample size has sufficient i
nformational value with respect to both statistical hypotheses linked to the
theory of Halperin and Schulz. Ultimately, examine this account in a communit
y sample can extend this ADHD theory to subclinical and general population.
```

Supplemental Text 2

Attention-deficit/hyperactivity disorder (ADHD) Symptom Distribution in our Study Compared to Normative Thresholds and Distributions

Procedure

We calculated the total scores (i.e., sum of the items) of ADHD symptoms for each of the three self-reports measures: the Barkley Adult ADHD Rating Scale-IV: Childhood symptoms (cBAARS-IV), the Barkley Adult ADHD Rating Scale-IV: Current symptoms (aBAARS-IV), and the Adult ADHD Self-Report Screening Scale for DSM The percentage of individuals scoring above the preliminary clinical cutoff was analyzed for each measure. For the cBAARS-IV and the aBAARS-IV, this cutoff was set to the 95th percentile, (Barkley, 2011) while for the ASRS-5, the cutoff was set to a direct score of 14 points (Ustun et al., 2017).

In addition, for the cBAARS and the aBAARS, we compared the distribution of the scores in our sample with an estimation of a Barkley's (2011) normative distribution obtained from a large representative sample of adults from the United States. Barkley reported the values corresponding to 22 and 20 percentiles for the cBAARS and the aBAARS, respectively. Based on the values of these percentiles, we generated a sample of 10,000 simulated total scores for the cBAARS and the aBAARS using a Monte Carlo simulation approach. Percentiles of the simulated sample fitted Barkley's percentiles for each ADHD symptom scale, and they may be compared with our study percentiles in Supplemental Table 3. Finally, we extracted 292 theoretical quantiles from the simulated normative sample, which resulted in a sample of equal size to our study sample. This equally sized sample was used for comparison purposes with the sample from our study.

Results

Binomial tests indicated that while the percentage of participants above the 95th percentile cutoff in the cBAARS-IV (7.5%) was not significantly different from the normative 5% ($p = .058$), such percentage in the aBAARS-IV (11.0%) was higher than that preliminary clinical cutoff ($p < .001$). Regarding the ASRS-5, the percentage of participants above the cutoff of 14 points (11.6%) was not statistically different from the normative 11.2% (p = 781).[10] Two-sample Kolmogorov-Smirnov tests indicate that ADHD symptom distributions in the cBAARS-IV and the aBAARS-IV were significatively different from their corresponding normative sample (both $p < .001$). Indeed, a visual comparison in Figure 3 shows that the normative distributions are clearly more left-skewed than their respective sample distributions. Compared to the 50th percentile in the normative sample for the cBAARS-IV (*Mdn* = 23) and the aBAARS-IV (*Mdn* = 22), Wilcoxon Signed-Ranks Tests indicated that the medians of the cBAARS-IV (*Mdn* = 29) and the aBAARS-IV (*Mdn* = 30) in our sample were significatively higher (both $p < .001$). Levene's tests for homogeneity of variances showed no significant differences between the variability of the scores of the cBAARS-IV and the aBAARS-IV between our sample and the normative one (both *p*s > .430).

[10] Note that the criterion of 11.2% comes from a sample of adults with 8.2% of ADHD cases (Ustun et al., 2017), which is a higher percentage than in the general population.

Supplemental Table 1

List of Deviations From the Preregistration

Deviation no.	Description
1	We preregistered that for participants in whom a period of more than 90 days elapsed from the initial survey until the completion of the task, we would use the ADHD symptom self-reports completed during the second phase of the study. However, at that time, we did not expect that seven of those participants (six valid) had not completed the questionnaires the second time. We decided to retain these participants for the preregistered analyses and add this decision point (i.e., retain vs. exclude) into the multiverse analyses. No changes in the pattern of statistical significances of the preregistered statistical hypotheses were found when these participants were excluded.
2	At the time of preregistration, we were not aware of the importance of controlling the inflation of the Type I error rate in secondary statistical hypotheses (Sc1 and Sc2). Since both refer to the same theoretical hypothesis as their corresponding primary statistical hypotheses (H1 and H2), in the final manuscript we decided to halve the statistical significance threshold for the secondary hypotheses (i.e., $\alpha/2 = .025$). The same correction was applied for the opposite counterparts of the secondary statistical hypotheses (Sc1′ and Sc2′).

Deviation no.	Description
3	We preregistered an exploratory analysis (E) that consisted of examining the mediating role of cognitive control in the decrement of vigilance associated with ADHD symptoms. However, since the association between ADHD symptoms and decrement of vigilance was smaller in size than the effects of interest, we decided not to conduct this mediation analysis.
4	As a sensitivity analysis to be included in the multiverse (Ss1), we preregistered an alternative approach to estimate late-developing ADHD symptoms. This consisted of computing the result of subtracting symptoms in childhood from symptoms in adulthood. However logical concerns with the use of change variables at different timepoints (Shahar & Shahar, 2012), along with a potential floor effect at least in the cBAARS-IV, made us realise the inadequacy of this approach. When change scores are computed for statistical hypotheses involving late-developing symptoms (i.e., H2, Sc2, H1′, Sc1′), the pattern of statistical significances is the same as in the preregistered analyses (i.e., all nonsignificant).
5	After the preregistration, we identified two decision points with more than one reasonable analytical option (for a more detailed rationale, see Supplemental Table 4): the correlation coefficient (Kendall vs. Spearman vs. Pearson) and the type of correlation (zero-order vs. partial). Both were included in the multiverse.

Note. ADHD = Attention-deficit/hyperactivity disorder; cBAARS-IV = Barkley Adult ADHD Rating Scale-IV: Childhood symptoms.

Supplemental Table 2

Effect Sizes of Interest and Achieved Power to Detect Them Across Each Opposite Statistical Hypothesis

Hypothesis	Smallest effect of interest			Expected effect (from Coll-Martín et al., 2021)		
	r	τ	Power (1 – β)	*r*	τ	Power (1 – β)
H1′	.13	**.08**	.67	–.09	–.06	0
H2′	–.15	**–.09**	.78	–.17	**–.11**	.88
Sc1′	.15	**.10**	.70	.05	.03	.11
Sc2′	–.18	**–.12**	.85	.09	.06	0

Note. $N = 292$. H1′ and H2′ are the opposite counterparts of the primary statistical hypotheses, while Sc1′ and Sc2′ are the opposite counterparts of the secondary statistical hypotheses. Kendall's τ values come from the Pearson's *r* values used to conduct the 10,000 simulations for power analysis (see Gilpin, 1993, for the formula to transform the correlation coefficients). Statistical power corresponds to one-side tests for Kendall's coefficients, with $\alpha = .05$ for opposite primary statistical hypotheses and $\alpha = .025$ for opposite secondary statistical hypotheses. Kendall's τ values above the minimal statistically detectable effect ($\tau = |.06|$ for opposite primary hypotheses; $\tau = |.08|$ for opposite secondary hypotheses) are in bold.

Supplemental Table 3

Percentiles Values of the cBAARS-IV and the aBAARS-IV in Original and Estimated Normative Samples Compared to the Sample of the Present Study

	cBAARS-IV total scores		aBAARS-IV total scores	
Percentile	Normative sample	Study sample	Normative sample	Study sample
99	60 (61)	58	54 (55)	53
98	55	54	49	50
97	52 (53)	52	46	50
96	51	52	44	48
95	49	51	43	48
94	47 (48)	49	42 (41)	47
93	46	49	39	45
92	45	47	38	44
91	44	47	37	43
90	43	46	36	43
89	42	46		
88	41 (40)	45		
87	39	45		
85	38	43	35	40
84	37	42		
83			34	39
82	36	41	33	38
81			32	38
79			31	37
77	35	38	30	37
76	34	38		
75	33	38	29	36
51	24 (23)	29	23 (22)	30
50	23	29	22	30
1	18	18	18	19

Note. Percentiles in the normative sample were extracted from Barkley (2011), which is based on a large representative sample of adults (ages 18–39) from the United States. These percentiles were used to generate a simulated normative sample through a Monte Carlo simulation approach (N = 10,000). The values of the simulated sample are in parentheses when they differ from the original normative sample. cBAARS-IV = Barkley Adult ADHD Rating Scale-IV: Childhood symptoms; aBAARS-IV = Barkley Adult ADHD Rating Scale-IV: Current symptoms.

Supplemental Table 4

Multiverse Analysis of the Correlations Between Arousal Vigilance (AV) and Executive Vigilance (EV) with ADHD Symptoms for Each Reasonable Combination of Analytical Decisions

Universe	Analytical decision					Correlation with ADHD symptoms		
	Long respondents	Poor task performers	Task index computation	Correlation coefficient	Correlation type	cBAARS-IV	ASRS-5	aBAARS-IV
	Primary vigilance outcome: Lapse slope (AV)							
1	**Retained**	**Excluded**	**Absolute**	**Kendall**	**Zero-order**	**.00**	–.00	.03
2	*Retained*	*Excluded*	*Absolute*	*Kendall*	*Partial*	.00	*.01*	.05
3	Retained	Excluded	Absolute	Spearman	Zero-order	.00	.00	.04
4	Retained	Excluded	Absolute	Spearman	Partial	.00	.01	.07
5	Retained	Excluded	Absolute	Pearson	Zero-order	.01	.02	.08
6	Retained	Excluded	Absolute	Pearson	Partial	.00	.02	.09
7	Retained	Excluded	Relative	Kendall	Zero-order	.00	.03	.06
8	Retained	Excluded	Relative	Kendall	Partial	–.02	.04	.08*
9	Retained	Excluded	Relative	Spearman	Zero-order	.00	.04	.08
10	Retained	Excluded	Relative	Spearman	Partial	–.03	.05	.12*
11	Retained	Excluded	Relative	Pearson	Zero-order	–.02	.04	.07
12	Retained	Excluded	Relative	Pearson	Partial	–.04	.05	.09
13	Retained	Retained	Absolute	Kendall	Zero-order	.01	.02	.04
14	Retained	Retained	Absolute	Kendall	Partial	.00	.01	.05
15	Retained	Retained	Absolute	Spearman	Zero-order	.01	.03	.06
16	Retained	Retained	Absolute	Spearman	Partial	.00	.02	.08
17	Retained	Retained	Absolute	Pearson	Zero-order	–.02	.03	.06
18	Retained	Retained	Absolute	Pearson	Partial	–.03	.04	.08
19	Retained	Retained	Relative	Kendall	Zero-order	–.00	.05	.07*
20	Retained	Retained	Relative	Kendall	Partial	–.03	.06	.10**
21	Retained	Retained	Relative	Spearman	Zero-order	–.00	.07	.10*
22	Retained	Retained	Relative	Spearman	Partial	–.05	.08	.15**
23	Retained	Retained	Relative	Pearson	Zero-order	–.05	.06	.06

Universe	Analytical decision					Correlation with ADHD symptoms		
	Long respondents	Poor task performers	Task index computation	Correlation coefficient	Correlation type	cBAARS-IV	ASRS-5	aBAARS-IV
				Primary vigilance outcome: Lapse slope (AV)				
24	Retained	Retained	Relative	Pearson	Partial	–.07	.08	.10*
25	Excluded	Excluded	Absolute	Kendall	Zero-order	.01	.01	.04
26	Excluded	Excluded	Absolute	Kendall	Partial	.01	.02	.05
27	Excluded	Excluded	Absolute	Spearman	Zero-order	.02	.02	.05
28	Excluded	Excluded	Absolute	Spearman	Partial	.01	.03	.08
29	Excluded	Excluded	Absolute	Pearson	Zero-order	.02	.04	.10
30	Excluded	Excluded	Absolute	Pearson	Partial	.00	.03	.10*
31	Excluded	Excluded	Relative	Kendall	Zero-order	.00	.04	.06
32	Excluded	Excluded	Relative	Kendall	Partial	–.02	.05	.09*
33	Excluded	Excluded	Relative	Spearman	Zero-order	.01	.05	.09
34	Excluded	Excluded	Relative	Spearman	Partial	–.03	.07	.13*
35	Excluded	Excluded	Relative	Pearson	Zero-order	–.02	.05	.07
36	Excluded	Excluded	Relative	Pearson	Partial	–.04	.06	.10*
37	Excluded	Retained	Absolute	Kendall	Zero-order	.01	.03	.05
38	Excluded	Retained	Absolute	Kendall	Partial	.00	.03	.06
39	Excluded	Retained	Absolute	Spearman	Zero-order	.02	.05	.08
40	Excluded	Retained	Absolute	Spearman	Partial	.01	.04	.09
41	Excluded	Retained	Absolute	Pearson	Zero-order	–.01	.07	.08
42	Excluded	Retained	Absolute	Pearson	Partial	–.03	.07	.10*
43	Excluded	Retained	Relative	Kendall	Zero-order	.00	.06	.08*
44	Excluded	Retained	Relative	Kendall	Partial	–.03	.07*	.11**
45	Excluded	Retained	Relative	Spearman	Zero-order	.00	.09	.11*
46	Excluded	Retained	Relative	Spearman	Partial	–.05	.10*	.16**
47	Excluded	Retained	Relative	Pearson	Zero-order	–.04	.09	.08
48	Excluded	Retained	Relative	Pearson	Partial	–.08	.11*	.12*
				Primary vigilance outcome: Hit slope (EV)				
1	*Retained*	*Excluded*	*Absolute*	*Kendall*	*Zero-order*	*.04*	–.01	.01
2	**Retained**	**Excluded**	**Absolute**	**Kendall**	**Partial**	.03	**–.03**	–.01
3	Retained	Excluded	Absolute	Spearman	Zero-order	.05	–.02	.02

Universe	Analytical decision					Correlation with ADHD symptoms		
	Long respondents	Poor task performers	Task index computation	Correlation coefficient	Correlation type	cBAARS-IV	ASRS-5	aBAARS-IV
	Primary vigilance outcome: Hit slope (EV)							
4	Retained	Excluded	Absolute	Spearman	Partial	.05	–.05	–.01
5	Retained	Excluded	Absolute	Pearson	Zero-order	.03	–.03	.00
6	Retained	Excluded	Absolute	Pearson	Partial	.04	–.04	–.01
7	Retained	Excluded	Relative	Kendall	Zero-order	.05	–.01	.02
8	Retained	Excluded	Relative	Kendall	Partial	.04	–.04	–.02
9	Retained	Excluded	Relative	Spearman	Zero-order	.07	–.02	.03
10	Retained	Excluded	Relative	Spearman	Partial	.06	–.06	–.02
11	Retained	Excluded	Relative	Pearson	Zero-order	.04	–.03	.00
12	Retained	Excluded	Relative	Pearson	Partial	.05	–.04	–.02
13	Retained	Retained	Absolute	Kendall	Zero-order	.04	–.01	.00
14	Retained	Retained	Absolute	Kendall	Partial	.04	–.04	–.02
15	Retained	Retained	Absolute	Spearman	Zero-order	.05	–.02	.01
16	Retained	Retained	Absolute	Spearman	Partial	.05	–.05	–.02
17	Retained	Retained	Absolute	Pearson	Zero-order	.04	–.03	–.01
18	Retained	Retained	Absolute	Pearson	Partial	.06	–.05	–.03
19	Retained	Retained	Relative	Kendall	Zero-order	.04	–.01	.01
20	Retained	Retained	Relative	Kendall	Partial	.05	–.04	–.03
21	Retained	Retained	Relative	Spearman	Zero-order	.06	–.02	.01
22	Retained	Retained	Relative	Spearman	Partial	.07	–.06	–.04
23	Retained	Retained	Relative	Pearson	Zero-order	.05	–.03	–.00
24	Retained	Retained	Relative	Pearson	Partial	.06	–.05	–.03
25	Excluded	Excluded	Absolute	Kendall	Zero-order	.03	–.02	.01
26	Excluded	Excluded	Absolute	Kendall	Partial	.03	–.03	–.01
27	Excluded	Excluded	Absolute	Spearman	Zero-order	.05	–.02	.01
28	Excluded	Excluded	Absolute	Spearman	Partial	.04	–.06	–.01
29	Excluded	Excluded	Absolute	Pearson	Zero-order	.02	–.04	–.01
30	Excluded	Excluded	Absolute	Pearson	Partial	.03	–.05	–.02
31	Excluded	Excluded	Relative	Kendall	Zero-order	.04	–.02	.01
32	Excluded	Excluded	Relative	Kendall	Partial	.04	–.04	–.02

Universe	Analytical decision					Correlation with ADHD symptoms		
	Long respondents	Poor task performers	Task index computation	Correlation coefficient	Correlation type	cBAARS-IV	ASRS-5	aBAARS-IV
				Primary vigilance outcome: Hit slope (EV)				
33	Excluded	Excluded	Relative	Spearman	Zero-order	.06	–.03	.02
34	Excluded	Excluded	Relative	Spearman	Partial	.06	–.06	–.03
35	Excluded	Excluded	Relative	Pearson	Zero-order	.03	–.04	–.00
36	Excluded	Excluded	Relative	Pearson	Partial	.05	–.05	–.02
37	Excluded	Retained	Absolute	Kendall	Zero-order	.03	–.02	–.00
38	Excluded	Retained	Absolute	Kendall	Partial	.03	–.04	–.03
39	Excluded	Retained	Absolute	Spearman	Zero-order	.05	–.03	–.00
40	Excluded	Retained	Absolute	Spearman	Partial	.05	–.07	–.03
41	Excluded	Retained	Absolute	Pearson	Zero-order	.03	–.05	–.02
42	Excluded	Retained	Absolute	Pearson	Partial	.05	–.07	–.04
43	Excluded	Retained	Relative	Kendall	Zero-order	.04	–.02	.00
44	Excluded	Retained	Relative	Kendall	Partial	.04	–.05	–.04
45	Excluded	Retained	Relative	Spearman	Zero-order	.06	–.04	.00
46	Excluded	Retained	Relative	Spearman	Partial	.06	–.07	–.05
47	Excluded	Retained	Relative	Pearson	Zero-order	.04	–.05	–.01
48	Excluded	Retained	Relative	Pearson	Partial	.06	–.07	–.04
				Secondary vigilance outcome: Lapse overall (AV)				
1	**Retained**	**Excluded**	**Absolute**	**Kendall**	**Zero-order**	**.07**	.00	.08*
2	*Retained*	*Excluded*	*Absolute*	*Kendall*	*Partial*	.06	*–.00*	.06
3	Retained	Excluded	Absolute	Spearman	Zero-order	.10	.01	.12*
4	Retained	Excluded	Absolute	Spearman	Partial	.08	.00	.10
5	Retained	Excluded	Absolute	Pearson	Zero-order	.18**	.17**	.21***
6	Retained	Excluded	Absolute	Pearson	Partial	.13*	.12*	.14*
7	Retained	Excluded	Relative	Kendall	Zero-order	–.02	.00	.01
8	Retained	Excluded	Relative	Kendall	Partial	–.03	.02	.03
9	Retained	Excluded	Relative	Spearman	Zero-order	–.04	.01	.01
10	Retained	Excluded	Relative	Spearman	Partial	–.04	.03	.04
11	Retained	Excluded	Relative	Pearson	Zero-order	–.03	.03	–.01
12	Retained	Excluded	Relative	Pearson	Partial	–.05	.04	.01

Universe	Analytical decision					Correlation with ADHD symptoms		
	Long respondents	Poor task performers	Task index computation	Correlation coefficient	Correlation type	cBAARS-IV	ASRS-5	aBAARS-IV
	Secondary vigilance outcome: Lapse overall (AV)							
13	Retained	Retained	Absolute	Kendall	Zero-order	.06	.01	.08*
14	Retained	Retained	Absolute	Kendall	Partial	.04	.01	.07
15	Retained	Retained	Absolute	Spearman	Zero-order	.09	.03	.12*
16	Retained	Retained	Absolute	Spearman	Partial	.06	.02	.11
17	Retained	Retained	Absolute	Pearson	Zero-order	.11	.13*	.16**
18	Retained	Retained	Absolute	Pearson	Partial	.07	.10	.12*
19	Retained	Retained	Relative	Kendall	Zero-order	–.03	.01	.01
20	Retained	Retained	Relative	Kendall	Partial	–.04	.02	.03
21	Retained	Retained	Relative	Spearman	Zero-order	–.05	.01	.01
22	Retained	Retained	Relative	Spearman	Partial	–.06	.03	.05
23	Retained	Retained	Relative	Pearson	Zero-order	–.08	–.04	–.05
24	Retained	Retained	Relative	Pearson	Partial	–.07	–.01	–.01
25	Excluded	Excluded	Absolute	Kendall	Zero-order	.06	–.00	.07
26	Excluded	Excluded	Absolute	Kendall	Partial	.05	–.01	.06
27	Excluded	Excluded	Absolute	Spearman	Zero-order	.09	.00	.11
28	Excluded	Excluded	Absolute	Spearman	Partial	.07	–.01	.09
29	Excluded	Excluded	Absolute	Pearson	Zero-order	.16**	.14*	.19**
30	Excluded	Excluded	Absolute	Pearson	Partial	.12*	.10	.13*
31	Excluded	Excluded	Relative	Kendall	Zero-order	–.03	–.00	.00
32	Excluded	Excluded	Relative	Kendall	Partial	–.04	.01	.02
33	Excluded	Excluded	Relative	Spearman	Zero-order	–.05	.00	.00
34	Excluded	Excluded	Relative	Spearman	Partial	–.05	.02	.03
35	Excluded	Excluded	Relative	Pearson	Zero-order	–.05	.02	–.02
36	Excluded	Excluded	Relative	Pearson	Partial	–.06	.03	.00
37	Excluded	Retained	Absolute	Kendall	Zero-order	.05	.02	.08*
38	Excluded	Retained	Absolute	Kendall	Partial	.03	.00	.07
39	Excluded	Retained	Absolute	Spearman	Zero-order	.08	.03	.12*
40	Excluded	Retained	Absolute	Spearman	Partial	.04	.01	.10
41	Excluded	Retained	Absolute	Pearson	Zero-order	.10	.12*	.15*

Universe	Analytical decision					Correlation with ADHD symptoms		
	Long respondents	Poor task performers	Task index computation	Correlation coefficient	Correlation type	cBAARS-IV	ASRS-5	aBAARS-IV
				Secondary vigilance outcome: Lapse overall (AV)				
42	Excluded	Retained	Absolute	Pearson	Partial	.06	.09	.11
43	Excluded	Retained	Relative	Kendall	Zero-order	–.04	.01	.01
44	Excluded	Retained	Relative	Kendall	Partial	–.05	.02	.03
45	Excluded	Retained	Relative	Spearman	Zero-order	–.06	.01	.01
46	Excluded	Retained	Relative	Spearman	Partial	–.07	.04	.05
47	Excluded	Retained	Relative	Pearson	Zero-order	–.08	–.03	–.05
48	Excluded	Retained	Relative	Pearson	Partial	–.08	.00	–.01
				Secondary vigilance outcome: Hit overall (EV)				
1	*Retained*	*Excluded*	*Absolute*	*Kendall*	*Zero-order*	*–.09**	–.07	–.11**
2	**Retained**	**Excluded**	**Absolute**	**Kendall**	**Partial**	–.08*	**–.04**	–.08*
3	Retained	Excluded	Absolute	Spearman	Zero-order	–.13*	–.10	–.15**
4	Retained	Excluded	Absolute	Spearman	Partial	–.12*	–.07	–.12*
5	Retained	Excluded	Absolute	Pearson	Zero-order	–.14*	–.15*	–.18**
6	Retained	Excluded	Absolute	Pearson	Partial	–.10	–.11	–.12*
7	Retained	Excluded	Relative	Kendall	Zero-order	–.09*	–.07	–.11**
8	Retained	Excluded	Relative	Kendall	Partial	–.08*	–.05	–.08*
9	Retained	Excluded	Relative	Spearman	Zero-order	–.13*	–.11	–.16**
10	Retained	Excluded	Relative	Spearman	Partial	–.12*	–.07	–.12*
11	Retained	Excluded	Relative	Pearson	Zero-order	–.14*	–.15**	–.18**
12	Retained	Excluded	Relative	Pearson	Partial	–.10	–.11	–.12*
13	Retained	Retained	Absolute	Kendall	Zero-order	–.07	–.07	–.11**
14	Retained	Retained	Absolute	Kendall	Partial	–.06	–.05	–.09*
15	Retained	Retained	Absolute	Spearman	Zero-order	–.11	–.11	–.16**
16	Retained	Retained	Absolute	Spearman	Partial	–.09	–.07	–.14*
17	Retained	Retained	Absolute	Pearson	Zero-order	–.10	–.13*	–.15**
18	Retained	Retained	Absolute	Pearson	Partial	–.06	–.10	–.12*
19	Retained	Retained	Relative	Kendall	Zero-order	–.08*	–.08	–.11**
20	Retained	Retained	Relative	Kendall	Partial	–.07	–.05	–.09*
21	Retained	Retained	Relative	Spearman	Zero-order	–.11	–.11	–.16**

Universe	Analytical decision					Correlation with ADHD symptoms		
	Long respondents	Poor task performers	Task index computation	Correlation coefficient	Correlation type	cBAARS-IV	ASRS-5	aBAARS-IV
	Secondary vigilance outcome: Hit overall (EV)							
22	Retained	Retained	Relative	Spearman	Partial	–.10	–.07	–.14*
23	Retained	Retained	Relative	Pearson	Zero-order	–.11	–.13*	–.16**
24	Retained	Retained	Relative	Pearson	Partial	–.06	–.10	–.12*
25	Excluded	Excluded	Absolute	Kendall	Zero-order	–.08*	–.07	–.11**
26	Excluded	Excluded	Absolute	Kendall	Partial	–.07	–.04	–.08*
27	Excluded	Excluded	Absolute	Spearman	Zero-order	–.12*	–.10	–.16**
28	Excluded	Excluded	Absolute	Spearman	Partial	–.11	–.06	–.12*
29	Excluded	Excluded	Absolute	Pearson	Zero-order	–.12*	–.12*	–.16**
30	Excluded	Excluded	Absolute	Pearson	Partial	–.09	–.08	–.11
31	Excluded	Excluded	Relative	Kendall	Zero-order	–.08*	–.07	–.11**
32	Excluded	Excluded	Relative	Kendall	Partial	–.08	–.04	–.09*
33	Excluded	Excluded	Relative	Spearman	Zero-order	–.12*	–.10	–.16**
34	Excluded	Excluded	Relative	Spearman	Partial	–.11	–.06	–.13*
35	Excluded	Excluded	Relative	Pearson	Zero-order	–.13*	–.12*	–.17**
36	Excluded	Excluded	Relative	Pearson	Partial	–.09	–.09	–.12*
37	Excluded	Retained	Absolute	Kendall	Zero-order	–.07	–.07	–.11**
38	Excluded	Retained	Absolute	Kendall	Partial	–.06	–.05	–.10*
39	Excluded	Retained	Absolute	Spearman	Zero-order	–.10	–.11	–.17**
40	Excluded	Retained	Absolute	Spearman	Partial	–.08	–.08	–.14*
41	Excluded	Retained	Absolute	Pearson	Zero-order	–.09	–.11	–.14*
42	Excluded	Retained	Absolute	Pearson	Partial	–.05	–.09	–.11
43	Excluded	Retained	Relative	Kendall	Zero-order	–.07	–.08*	–.12**
44	Excluded	Retained	Relative	Kendall	Partial	–.06	–.05	–.10*
45	Excluded	Retained	Relative	Spearman	Zero-order	–.10	–.11	–.17**
46	Excluded	Retained	Relative	Spearman	Partial	–.08	–.08	–.15*
47	Excluded	Retained	Relative	Pearson	Zero-order	–.09	–.12*	–.15*
48	Excluded	Retained	Relative	Pearson	Partial	–.06	–.09	–.12*

Note. $N = 292$. Long respondents ($n = 7$; $n = 6$ after excluding poor task performers) are participants for whom more than 90 days elapsed between the completion of the ADHD self-reports of the task and the task itself. These participants did not complete the second administration of ADHD questionnaires (but note that the measures of ADHD symptoms in adulthood considers the last 6 months (i.e., twice our cut-off point). Poor task performers ($n = 14$; $n = 13$ after excluding long respondents) are participants with more than 25% errors in Attention Network Test for Interaction (ANTI) trials of the cognitive task. Note that the options of long respondents and poor performers slightly affect the sample size and therefore the statistical power. We control for this statistical confounder by directly inputting a sample size of 292 to the formula to obtain the p value from the correlation coefficient. The task index computation can be based on an absolute (i.e., reaction time [RT] higher than 600 ms or no response for lapses; correct detection of the displacement for hits) or relative (i.e., RT higher than the participant's mean plus one standard deviation or no response for lapses outcomes; correct detection of the displacement within 2.5 standard deviations plus/minus each participant's mean RT) threshold. Both poor task performers and task index computation were preregistered with the label (Ss3). The correlation coefficients include the nonparametric Kendall's rank-order correlation coefficient and Spearman rank-order correlation, but also the parametric Pearson product–moment correlation coefficient. Although the correlation of the pre-registered hypotheses violated the normalisation assumption, McDonald (2014) recommends using the Pearson correlation even in this type of scenario. In partial correlations, the associations involving the cBAARS-IV are controlled for the ASRS-5, while correlations involving the ASRS-5 or the aBAARS-IV are accounted for by the cBAARS-IV. Note that controlling for symptoms in childhood by a later variable (i.e., symptoms in adulthood) may be considered problematic by some due to potential collision bias (Rohrer, 2018). The rows with the set of

preregistered analytical decisions for each statistical hypothesis derived from Halperin and Schulz's (2006) neurodevelopmental model, along with their corresponding correlation, are in bold. The rows with the set of preregistered analytical decisions for each opposite statistical hypothesis (i.e., hypotheses relating each vigilance component to ADHD symptom severity in the age period opposite to that established in the neurodevelopmental model), along with their corresponding correlation, are italicised. ADHD = Attention-deficit/hyperactivity disorder; cBAARS-IV = Barkley Adult ADHD Rating Scale-IV: Childhood symptoms; aBAARS-IV = Barkley Adult ADHD Rating Scale-IV: Current symptoms; ASRS-5 = Adult ADHD Self-Report Screening Scale for DSM-5. $^{*}p < .05$, one-tailed. $^{**}p < .01$, one-tailed. $^{*}p < .001$, one-tailed. All contrasts are corrected for multiple comparisons in hypotheses involving secondary vigilance outcomes.

Supplemental Table 5

Means, Standard Deviations, and Pearson Correlations Among Arousal and Executive ANTI-Vea Indices

Index	*M*	*SD*	1	2	3	4	5	6
Arousal								
1. Lapse slope (AV)	1.09	2.60	—					
2. Lapse overall (AV)	10.57	17.69	.22***	—				
3. RT *M* slope (AV)	2.07	10.12	.77***	.21***	—			
4. RT *M* overall (AV)	491.75	95.44	.18**	.88***	.15**	—		
5. RT *SD* slope (AV)	4.33	9.41	.54***	.26***	.49***	.23***	—	
6. RT *SD* overall (AV)	78.73	32.79	.37***	.63***	.35***	.51***	.44***	—
7. Alerting RT	37.20	31.60	.07	–.03	.06	.03	.02	–.02
8. Alerting errors	2.46	4.60	.18**	.08	.13*	.10	.14*	.15*
Executive								
9. Hits slope (EV)	–2.17	3.78	–.10	–.01	–.08	.02	–.12*	–.11
10. Hits overall (EV)	73.08	18.28	–.07	–.11	–.06	–.11	–.07	–.18**
11. FA slope (EV) [a]	–0.51	1.71	–.01	.04	–.08	.05	.01	.08
12. FA overall (EV) [a]	5.41	5.19	.02	.09	.07	.10	.06	.07
13. Congruency RT	42.54	27.71	.07	.09	.02	.09	.12*	.16**
14. Congruency errors	0.92	4.18	.13*	–.08	.03	–.09	.07	.07

Index	7	8	9	10	11	12	13	14
Arousal								
1. Lapse slope (AV)								
2. Lapse overall (AV)								
3. RT *M* slope (AV)								
4. RT *M* overall (AV)								
5. RT *SD* slope (AV)								
6. RT *SD* overall (AV)								
7. Alerting RT	—							
8. Alerting errors	.05	—						
Executive								
9. Hits slope (EV)	.03	.02	—					
10. Hits overall (EV)	.12*	.03	.41***	—				
11. FA slope (EV) [a]	.01	–.02	.07	-.04	—			
12. FA overall (EV) [a]	.22***	–.01	.21***	.27***	–.21***	—		
13. Congruency RT	–.10	.07	–.13*	–.31***	.10	–.22***	—	
14. Congruency errors	–.10	.05	–.11	–.45***	.14*	–.32***	.35***	—

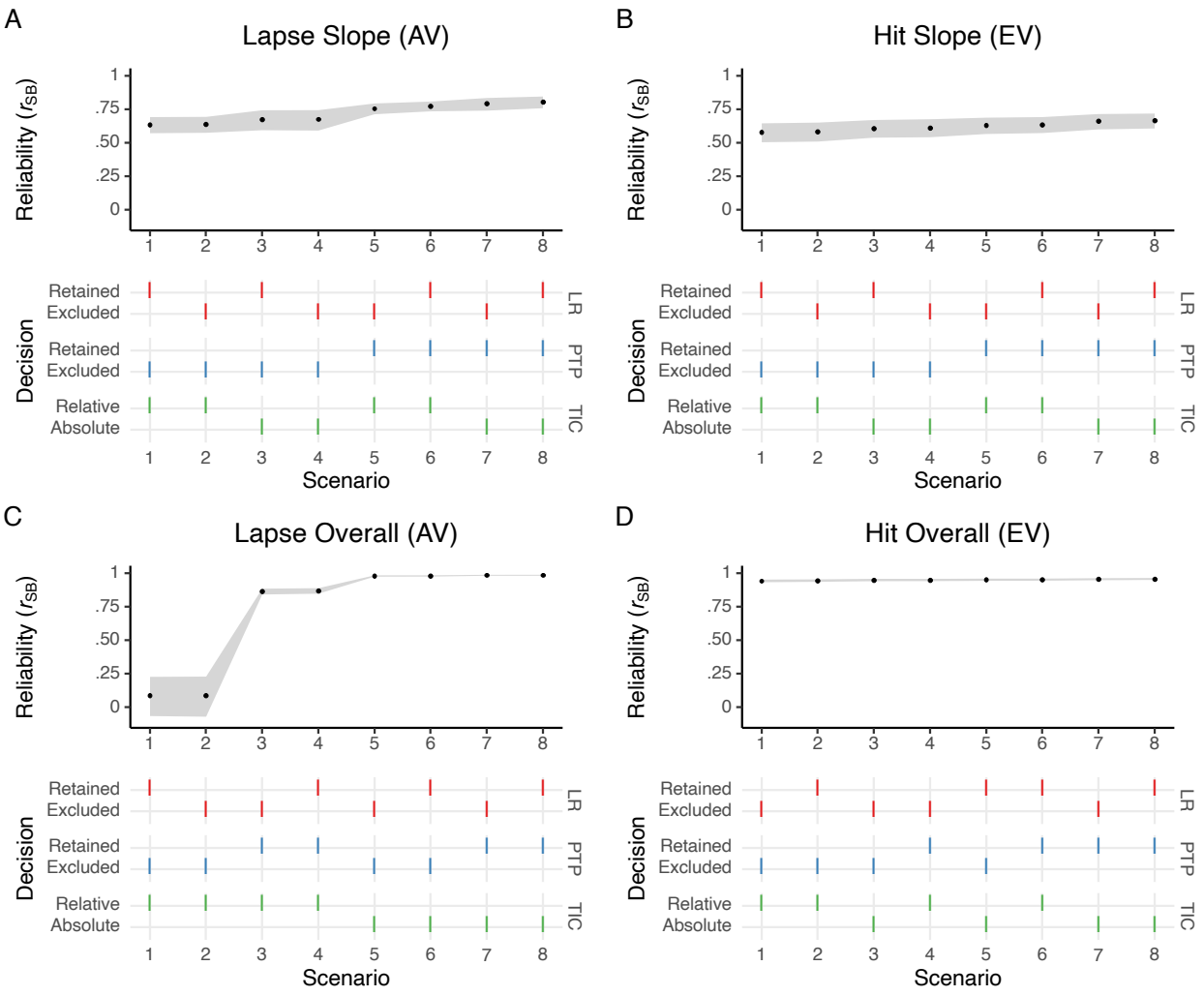
Note. $N = 292$. These indices come from the Analysis section of the website at https://anti-vea.ugr.es/analysis.html (for a more detailed description of the analyses, see Luna et al., 2020). ANTI-Vea = Attention Network Test for Interactions and Vigilance—executive and arousal components; RT = Reaction time; AV =Arousal vigilance; EV = Executive vigilance; FA = False alarm.

[a] $n = 289$.

$^{*}p < .05$. $^{**}p < .01$. $^{***}p < .001$.

Supplemental Figure 1

Internal Consistency Reliability Scores for Indices of Arousal Vigilance (AV) and Executive Vigilance (EV) Across the Multiverse of Reasonable Analytical Options



Note. $N = 292$ (LR retained and PTP excluded; as preregistered); $N = 306$ (LR retained and PTP retained); $N = 299$ (LR excluded and PTP retained); $N = 286$ (LR excluded and PTP excluded). Internal consistency scores were estimated using a permutation-based split-half approach (Parsons et al., 2019) with 10,000 random splits, and then applying the Spearman–Brown prophecy formula. The black dots represent the median scores of each set of iterations, while the grey area represents their corresponding 95% confidence interval (based on 2.5th and 97.5th

percentiles of each set of iterations). r_{SB} = Spearman-Brown split-half reliability coefficient; LR = Long respondents; PTP = Poor task performers; TIC = Task index computation.

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